



INTERNATIONAL ENERGY AGENCY

Implementing Agreement for Co-operation in the Research,
Development and Deployment of Wind Turbine Systems
Task 11

Topical Expert Meeting #84 on

Aerodynamics of Wind Turbines

IEA Joint Action Task 11- Task 29 Topical expert meeting on aerodynamics

January 11-13, 2016

NWTC NREL, Boulder, USA



CENER

NATIONAL RENEWABLE
ENERGY CENTRE

Scientific Coordination: Xabier Munduate and Oscar.Pires

CENER (Centro Nacional de Energías Renovables)

Ciudad de la Innovación 7, 31621 Sarriguren (Navarra) Spain

Phone: +34 948 25 28 00

E-mail: xmunduate@cener.com

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CENER
Xabier Munduate
Ciudad de la Innovación 7
31621 Sarriguren (Navarra) Spain
Phone: +34 948 25 28 00
E-mail: xmunduate@cener.com

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International Energy Agency

Implement Agreement for Co-operation in the Research, Development and Deployment of Wind Turbine Systems: IEA Wind

The IEA international collaboration on energy technology and RD&D is organized under the legal structure of Implementing Agreements, in which Governments, or their delegated agents, participate as Contracting Parties and undertake Tasks identified in specific Annexes.

The IEA's Wind Implementing Agreement began in 1977, and is now called the Implementing Agreement for Co-operation in the Research, Development, and Deployment of Wind Energy Systems (IEA Wind). At present, 24 contracting parties from 20 countries, the European Commission, and the European Wind Energy Association (EWEA) participate in IEA Wind. Austria, Canada, Denmark, the European Commission, EWEA, France, Finland, Germany, Greece, Ireland, Italy (two contracting parties), Japan, Republic of Korea, Mexico, Netherlands, Norway (two contracting parties), Portugal, Spain, Sweden, Switzerland, United Kingdom and the United States are now members.

The development and maturing of wind energy technology over the past 30 years has been facilitated through vigorous national programs of research, development, demonstration, and financial incentives. In this process, IEA Wind has played a role by providing a flexible framework for cost-effective joint research projects and information exchange.

The mission of the IEA Wind Agreement continues to be to encourage and support the technological development and global deployment of wind energy technology. To do this, the contracting parties exchange information on their continuing and planned activities and participate in IEA Wind Tasks regarding cooperative research, development, and demonstration of wind systems.

Task 11 of the IEA Wind Agreement, Base Technology Information Exchange, has the objective to promote and disseminate knowledge through cooperative activities and information exchange on R&D topics of common interest to the Task members. These cooperative activities have been part of the Wind Implementing Agreement since 1978.

Task 11 is an important instrument of IEA Wind. It can react flexibly on new technical and scientific developments and information needs. It brings the latest knowledge to wind energy players in the member countries and collects information and recommendations for the work of the IEA Wind Agreement. Task 11 is also an important catalyst for starting new tasks within IEA Wind.

IEA Wind TASK 11: BASE TECHNOLOGY INFORMATION EXCHANGE

The objective of this Task is to promote disseminating knowledge through cooperative activities and information exchange on R&D topics of common interest. Four meetings on different topics are arranged every year, gathering active researchers and experts. These cooperative activities have been part of the Agreement since 1978.



Carballeira Wind Farm - Spain

Two Subtasks

The task includes two subtasks.

The objective of the first subtask is to develop recommended practices (RP). In 2013 were edited RPs on “Social Acceptance of Wind Energy Projects”, “Wind Integration Studies” and. “Ground-Based Vertically Profiling Remote Sensing for Wind Resource Assessment”.

The objective of the second subtask is to conduct topical expert meetings in research areas identified by the IEA R&D Wind Executive Committee. The Executive Committee designates topics in research areas of current interest, which requires an exchange of information. So far, Topical Expert Meetings are arranged four times a year.

Documentation

Since these activities were initiated in 1978, more than 70 volumes of proceedings have been published. In the series of Recommended Practices 16 documents were published and five of these have revised editions.

All documents produced under Task 11 and published by the Operating Agent are available to citizens of member countries participating in this Task.

Operating Agent

CENER
 Xabier Munduate
 Ciudad de la Innovación 7
 31621 Sarriena (Navarra) Spain
 Phone: +34 948 25 28 00
 E-mail: xmunduate@cener.com

COUNTRIES PRESENTLY PARTICIPATING IN THE TASK 11	
COUNTRY	INSTITUTION
Denmark	Danish Technical University (DTU) - Risø National Laboratory
Republic of China	Chinese Wind Energy Association (CWEA)
Finland	Technical Research Centre of Finland - VTT Energy
Germany	Bundesministerium für Umwelt , Naturschutz und Reaktorsicherheit -BMU
Ireland	Sustainable Energy Ireland - SEI
Italy	Ricerca sul sistema energetico, (RSE S.p.A.)
Japan	National Institute of Advanced Industrial Science and Technology AIST
Mexico	Instituto de Investigaciones Electricas - IEE
Netherlands	Rijksdienst voor Ondernemend Nederland (RVO)
Norway	The Norwegian Water Resources and Energy Directorate - NVE
Spain	Centro de Investigaciones Energéticas, Medioambientales y Tecnológicas CIEMAT
Sweden	Energimyndigheten – Swedish Energy Agency
Switzerland	Swiss Federal Office of Energy - SFOE
United Kingdom	CATAPULT Offshore Renewable Energy
United States	The U.S Department of Energy -DOE

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1. INTRODUCTORY NOTE

This meeting belongs to the series of Joint Action Symposiums on Aerodynamics for wind turbines and took place at the NWTC of NREL, Boulder, Colorado, USA. The meeting gathered 16 participants representing 7 different countries. The meeting was held on January 13th immediately after the meeting of the IEA Task 29, 11th-12th January.

Expected Outcome

The goal of the meeting is to gather knowledge on recent aerodynamic numerical and experimental techniques and developments to make maximum growth of wind energy. By gathering and exchanging information we hope to achieve a common understanding of the aerodynamic issues that affect wind turbines.

These issues are the way wind turbines receive the inflow, how the rotor blades see the pressure and friction acting on the surface of the blades and how the wake develops and influences the rotor aerodynamic induction. It was a “freewheeling meeting” where aerodynamic experts all over the world are invited to present their work on an open aerodynamic subject. This was set up together with the coordinator of Task 29 as was approved from last year IEA ExCo meeting.

2. AGENDA

Wednesday 13th January

Morning presentations 9:00am-12:15am

Registration. Collection of presentations

Introduction by Host

Scott Schreck, NWTC - NREL, USA

Introduction by Task 11 Operating Agent

Xabier Munduate, CENER

1- Niels Nørmark Sørensen, Technical University of Denmark, Prediction of airfoil performance at high Reynolds numbers.

2.- Christoph Heister, German Aerospace Center (DLR), CFD simulations of the MexNext rotor experiment using the DLR TAU code.

3.- James Brasseur, University of Colorado, Analysis of Nonsteady Transitional Boundary layer Dynamics for Flow over an Oscillating Wind Turbine Airfoil.

4.- Alois Peter Schaffarczyk, Kiel University of Applied Sciences, Remarks on the Life of Ludwig Prandtl (1875 – 1953).

5.- Xabier Munduate, CENER, Finite Number of Blades Effect on the Tip and the Root.

Working Lunch 12:15 1:15

Introduction by Daniel Laird, Director of the National Wind Technology Center, on the R&D programs and opportunities for collaboration between USA and other international partners.

Afternoon presentations 1:15 4:30

6.-Helge Aagaard Madsen, Technical University of Denmark, Wake flow characteristics at high wind speed.

7.- Koen Boorsma, ECN, -Application of a vortex line method for dynamic load case calculations.

Coffee break 15 minutes

8. Srinivas Guntur, NWTC NREL, Using Siemens 2.3MW experimental data for FAST verification and validation.

9.- Gerard Schepers, ECN, General overview of Avatar project.

4:30pm 5:00pm Summary of Aerodynamics of Wind Turbines

3. LIST OF PARTICIPANTS

The meeting was attended by 16 participants from 7 countries. Following, the lists of participants and their affiliations.

Ye Zhang	Delft University of Technology
Alois Peter Schaffarczyk	Kiel University of Applied Sciences
Luca Oggiano	IFE
Koen Boorsma	ECN
Gerard Schepers	ECN
Christoph Heister	German Aerospace Center (DLR)
Helge Aagaard Madsen	Technical University of Denmark
Niels Nørmark Sørensen	Technical University of Denmark
Simon-Philippe Breton	Uppsala University/Department of Earth Science
Xabier Munduate Echarri	CENER
Srinivas Guntur	NREL
Lee Jay Fingersh	NREL
Jason Jonkman	NREL
Scot Schreck	NREL
Dick Veldkamp	Suzlon
Jim Brasseur	University of Colorado

4. SUMMARY

In brevity the following topics were presented and discussed.

- **Niels N. Sørensen**, Technical University of Denmark, “Prediction of airfoil performance at high Reynolds numbers.” Increasing the rotor size leads to high Mach numbers and high Reynolds numbers. Here in only the later will be addressed through a study of the two transitions models to predict the change from laminar to turbulent flow: The γ - Re_θ Correlation based transition model and the e^n model for natural transition. Reynolds number from [3-40] million are computed with DTU, EllipSys2D incompressible solver. After evaluating the two transition models with three test cases airfoils the γ - Re_θ correlation based model do not respond correctly to varying Reynolds number, while the e^n model for natural transition behaves better. The conclusions are that since wind turbine rotors will face high Reynolds with increasing size:

-The available data show that the γ - Re_θ model over-predicts drag at high Reynolds

-Very little wind tunnel experimental data available for comparison.

- **Christoph Heister**, German Aerospace Center (DLR), “CFD simulations of the MexNext rotor experiment using the DLR TAU code”. An overview of MexNext experimental test cases followed by CFD simulations using the DLR TAU RANS code are presented together with results from the validation performed. The simulation setup is an isolated rotor without the hub-nacelle geometry. The trajectory of tip vortex can be identified and how the root vortex develops in the CFD computations, as well as the axial, azimuthal and radial transverse velocities, pressure distributions on the blade sections and integrated torque and axial force.

The conclusion is that the TAU RANS results show physically feasible description of 3D rotor flow field. Further work is required related to:

- TAU simulations with tripped/free transition and addition of further wind turbine components (e.g. nacelle/spinner)

- Unsteady TAU simulation of Case 1.3 (stalled rotor)

- **James G. Brasseur**, University of Colorado, “Analysis of Nonsteady Transitional Boundary layer Dynamics for Flow over an Oscillating Wind Turbine Airfoil using the γ - Re_θ Transition Model”. Nonsteady loading response to passage of energetic atmospheric turbulence eddies, and consequences in the unsteady aerodynamics of the airfoil and blade are presented. The study covers all the airflow scales, from Mesoscale Weather (URANS

WRF) through Atmospheric Boundary Layer and Turbulence Eddies (4D LES) to Blade Aerodynamics (URANS-LES and AL).

There are load and shaft-moment fluctuations correlated with multiple atmospheric eddy passages. The rotor torque fluctuates and the central role of nonsteady changes in the Angle of Attack (AoA) is observed, e.g, the root part AoA changes from 5deg to 15deg.

He shows a validation of experiment datasets oscillating airfoil experiments (S809: Ohio State, Glasgow, Delft) against computer simulations using: OpenFOAM URANS and the γ - Re_θ transition model. The previous step has been to carry out CFD simulations of turbine blade in atmospheric boundary layer to estimate the three no dimensional parameters that characterize the degree of Hysteresis for the nonsteady changes in angle of attack (AoA). The time variations in AoA from the rotation of wind turbine blade through atmospheric eddies creates strong time variations in blade loadings over a large percentage of the blade. The central role of the transition process in the physics of the dynamic stall phenomena is observed. The conclusion from the work is:

-A transition model is required to predict nonsteady behaviours associated with separation at high AoAs and unsteady changes in angle of attack (AoA) and dynamic stall.

- **Alois Peter Schaffarczyk**, Kiel University of Applied Sciences, “Remarks on the Life of Ludwig Prandtl (1875 – 1953)”. The content of the talk goes into a retrospective study of the physics, engineering and historical remarks on the life of Prandtl. During his life, 1875-1953 he supervised around 85 PhD students. Some of his first work was on solid mechanics and plasticity, before addressing aerodynamics. One major contribution to the field of fluid dynamics was the creation of his Boundary Layer Theory 1904, overcoming d’Alembert’s Paradoxon with the viscous drag. He continued into the aerodynamic field contributing with the Lifting Line Theory and via Betz work on screw propellers with lowest resistance, Prandtl added an annex to Betz work, related to wind turbine aerodynamics, the effect of the number of blades (The famous tip correction). Progress on turbulence, Prandtl’s mixing length approach is an adaption from kinetic theory of gases to lumps of turbulence. Successful to jets and still in use for LES. Less known is that Prandtl (and Kolmogorov) published both in 1945 one (two)-differential-equation models of turbulence. From a historical perspective the Nazi years (1933 – 1945) they have also influenced research and scientific teams. In his (Prandtl) own statement from 1946: he considers himself as a “Mitläufer“ (follower) of Nazis. He never got the Nobel-Prize, for his Boundary Layer, Lifting line or Turbulence work, but he got nominated for the Nobel in 1935 for his work on turbulence.

Summary

- Outstanding Contributions to Aerodynamics
- Close but 2nd to Nobel-Prize level
- Benefitted from close connections to Nazi-leader

- **Xabier Munduate**, CENER, “Finite Number of Blades, Effect on the tip and the root”.

The effect of the finite number of blades (Tip loss effect) has been study in a novel approach, via a simple vortex model of the rotor. It follows the Joukowsky constant circulation concept with infinite rectilinear filaments of circulation at the tip and the root of each blade. The circulation (the tangential induced velocity V_θ) just behind the rotor, promoted by the root and tip rectilinear vortex filaments, has been calculated all along the span of the blade. This tangential induction, has been obtained on rotors with 1,2,3 and virtually infinite blades (100) resulting in:

- The circumferential axisymmetry of the flow, no variation in azimuth of the tangential induction, for the infinite number (100) of blades case has been obtained.
- The main result is the appearance of the non axisymmetry of the flow, azimuthal variation of the tangential induction over a rotation from root to the tip for the finite number of blades 1,2,3 rotor.

A method to obtain the azimuthal variation of the tangential induction is presented. This work concludes in a new manner of obtaining the tip loss effects.

- **Helge A. Madsen**, Technical University of Denmark, “Wake flow characteristics at high wind speed”. A sequential, engineering model approach for computation of the flow aligned with a row of turbines has been used to study the wake flow characteristics, below as well as above rated power. The experimental load data from the Lillgrund Wind Farm and the aeroelastic data input for the SWT 2.3 MW PRVS turbine hav been used. The motivation for study is the results from comparisons of measurements at the Lillgrund wind farm with simulations with the DWM, Dynamic Wake Method, which considers: Model Velocity deficit, Wake meandering and Wake added turbulence. Now the merging of wakes in the DWM model has changed. Two Wake merging methods are presented, the SUM method and the MIN method. Apparently the SUM method is more promising

Conclusions

- The simulations show that the last turbines in a row can experience inflow with a very strong velocity deficit that causes high loading.
- A so-called non-dimensional equilibrium wind speed is found at some distance down in the row, depending mainly on spacing and probably ambient turbulence intensity.
- The simulation results show that a summation of the deficits at high wind speed is a good choice, although still deviating for decreasing, increasing wind speed.

- **Koen Boorsma**, ECN, “Application of a vortex line method for dynamic load case calculations”. A comparison of experimental data, ECN BEM code and the ECN Vortex line AWSM code are presented for yawed flow, extreme transient shear and half wake. The ECN-AWSM code is compound of a vortex line for the computation of the blade aerodynamics plus a choice of free, fixed, prescribed or hybrid approach for simulating the wake. When compare the BEM and the vortex line method, for a Yawed flow (30°), results showed small differences in terms of loads but large differences in terms of induced velocities. Results for extreme transient shear and half wake are in good agreement.

As for the conclusions and future work:

- AWSM vortex line modelling often damps fluctuation in case of large dynamics.
- Note difference between BEM codes is often of same order.
- Improve AWSM vortex line CPU performance.

- **Srinivas Guntur**, NWTN NREL, “Using Siemens 2.3MW experimental data for FAST verification and validation”.

The objective of the work is to highlight the improvements of the latest version of FAST v8 over its ancestor versions. He compares results from FAST v8 simulations with experimental data as well as with the results from Siemens in-house aeroelastic simulation tool BHawC.

He uses Siemens 2.3MW 108-m-diameter rotor turbine experimental data to perform the validation. The improvements to examine in FAST v8 are, from one side the change from a simpler structural model (ElastoDyn) compared to new non-linear high-fidelity beam model (BeamDyn). On the other hand, the straight aerodynamic blades (AeroDyn v14) compared to aerodynamic blades curved in the 3D space (AeroDyn v15). The results of power curve, blade-tip deflections, blade-root and main shaft bending moments and tower-bottom fore-aft moment are in good correlation. The three-way comparison is performed between results from FAST v8, results from Siemens BHawC, and the experimental measurements. As a more detailed conclusion of the results it may be highlighted:

- Overall, FAST with BeamDyn shows significant improvement over ElastoDyn
- Improvement is primarily in the blade-tip deflection, but this improvement also propagates into other quantities of interest.
- Use of straight vs curved aerodynamic blade shows no significant impact.
- The experimental data is generally of very high quality.
- Minor discrepancies between FAST and Siemens’ data may be related to the controller or the shear model.

- **Gerard Schepers**, ECN, General overview of Avatar project.

It is EU FP7 project initiated by EERA, European Energy Research Alliance, where 11 universities and research centres plus 2 industrial partners are working since November 1st 2013 to November 1st 2017 with the main motivation of Aerodynamics of large wind turbines (10-20MW). The underplaying problem is that 10MW+ rotors violate assumptions in current aerodynamic tools, e.g.: Reynolds number effects, Compressibility effects, Flow transition and separation, (More) flexible blades, Flow devices. Hence 10MW+ designs fall outside the validated range of the current state of the art tools. The objective of the project is to bring the aerodynamic models to a next level and calibrate them for all relevant aspects of large (10MW+) wind turbines. As nowadays, no 10 MW turbines are on the market yet for validation, hence the way is to validate submodels against experiments. Several experimental data have been used:

- Measurements in a pressurized wind tunnel
- Measurements taken at controlled turbulence
- Measurements taken with flow devices

Models are also evaluated on two Reference Wind Turbines

- AVATAR low induction turbine with special aerodynamic challenges
- INNWIND.EU conventional induction turbine

The Work Package structure of the complete project, coordinated by ECN is:

- WP1: Integration and evaluation of the 10MW Reference Wind Turbine (CRES)
- WP2: Advanced aerodynamic modelling (DTU)
- WP3: Modelling of flow devices and flow control (CENER)
- WP4: Aeroelastic analysis of large flexible blades (NTUA)

The results from the project are available at

- <http://www.eera-avatar.eu/publications-results-and-links/>

Summary of Aerodynamics of Wind Turbines

Some participants expressed that it is difficult to compare aerodynamic results of wind turbines. Questions to be answered are for example how similar are the inputs to the modelling and how to measure and quantify the results from the different codes. The predictive tools will benefit from further research in collaborative and focus orientated projects. The general understanding was, however, that improved models will be important to better predict aerodynamics of wind turbines. Codes benchmarking is showing the importance of aerodynamics research. There are ongoing international efforts in verification/validation projects, where a number of computer simulations are being compared incorporating different level of refinements regarding aerodynamic effects.

On the subject of further aerodynamic research the list below shows the intentions of the participant's future research:

High Reynolds effects on transition, separation and performance.
CFD methods, especially transition models.
Unsteady aerodynamics and Dynamic stall modelling in ABL conditions.
Tip loss effects.
Develop Vortex codes and Dynamic Wake models.
Aeroelastic improvements and validation of engineering methods.

PRESENTATIONS



Prediction of airfoil performance at high Reynolds numbers

EFMC 2014, Copenhagen
17-20 Sept 2014

Niels N. Sørensen¹, F. Zahle¹, J.A. Michelsen²

¹ Wind Energy Department - DTU, Risø Campus, DK

²Department of Fluid Mechanics, DTU

DTU Wind Energy
Department of Wind Energy



Introduction

Large Scale Wind Turbines

Increasing the rotor size may potentially lead to two obvious aerodynamic issues

- ◆ High Mach numbers in the tip region
 - ◆ Possible to avoid
 - ◆ Might be harmful for performance
- ◆ High Reynolds numbers
 - ◆ Hard to avoid
 - ◆ Might be beneficial for performance
 - ◆ Difficult to measure in controlled environment



DTU 10 MW Reference Turbine



Introduction

Airfoil performance at high Reynolds Numbers

We expect that increasing the Reynolds Number will:

- ◆ Decrease the viscous effects due to the thinning of the boundary layer
- ◆ Promote earlier transition due to increased Reynolds number

Quantification the effects can be done by:

- ◆ Measurements
 - ◆ Tunnel measurements are difficult to obtain at high Re and low M
 - ◆ Openly available data are sparse
- ◆ Computations
 - ◆ Model performance in this range is unknown

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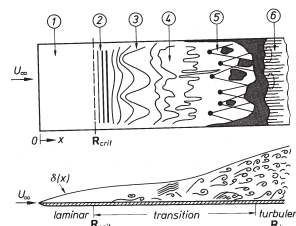
Prediction of airfoil performance at high Reynolds numbers



Introduction

Laminar turbulent transition

- ◆ The transition process depend on many parameters
 - ◆ Reynolds Number
 - ◆ Free stream turbulence level
 - ◆ Laminar separation bubbles
 - ◆ Cross flow
 - ◆ Surface roughness
 - ◆ Mass injection
- ◆ Typically approaches for transition modeling
 - ◆ e^n method (Orr-Sommerfeld eqn.)
 - ◆ Empirical correlations
 - ◆ Michel
 - ◆ Mayle
 - ◆ Abu-Ghannam and Shaw
 - ◆ Suzen



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Prediction of airfoil performance at high Reynolds numbers



Introduction

The $\gamma - Re_\theta$ Correlation based transition model

- ◆ The model is based on comparing the local Momentum Thickness Reynolds number with a critical value from empirical expressions

$$Re_\theta = Re_{\theta t}$$

- ◆ The following relation is used to simplify the computations in a general CFD code

$$Re_\theta = \frac{Re_{\nu t max}}{2.193}$$

- ◆ The model is based on transport equations, and can easily be implemented in general purpose flow solvers
- ◆ In the present form the model handles natural transition, by-pass transition, and separation induced transition
- ◆ The transition model is coupled to the $k - \omega$ SST model through the production and destruction terms in the k-equation

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Prediction of airfoil performance at high Reynolds numbers



Introduction

E^n model for natural transition

The E^n method is based on analyzing the behavior of small disturbances in the boundary layer

$$\psi(y) = \phi(y) \exp[i(\alpha x - \omega t)]$$

The disturbances are inserted in the Navier-Stokes equations, and linearized to give the Orr-Sommerfeld equation

$$(U^* - c^*)(\phi'' - \alpha^2 \phi) - (u^*)'' \phi = \frac{-i}{\alpha Re_\theta} (\phi'''' - 2\alpha^2 \phi'' + \alpha^4 \phi)$$

The Orr-Sommerfeld equation is solved with the following boundary conditions:

$$\phi(0) = \phi'(0) = 0 \text{ and } \phi(\infty) = \phi'(\infty) = 0$$

In the EllipSys, the E^n model can be used together with a bypass and a bubble criteria.

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Prediction of airfoil performance at high Reynolds numbers

Computational Setup Flow Solver



- ◆ We use the EllipSys2D incompressible solver.
- ◆ Diffusive terms by second order accurate central differences.
- ◆ Convective terms by QUICK.
- ◆ Steady state computations.
- ◆ Turbulence modeling by the $k - \omega$ SST model
- ◆ Transitional computations using $\gamma - Re_{\theta t}$ transition model and E^n model
- ◆ Reynolds number [3-40] million.

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Prediction of airfoil performance at high Reynolds numbers

Test Cases Selected cases



- ◆ NACA63 – 018
- ◆ DU00 – W – 212
- ◆ NACA64₂A015

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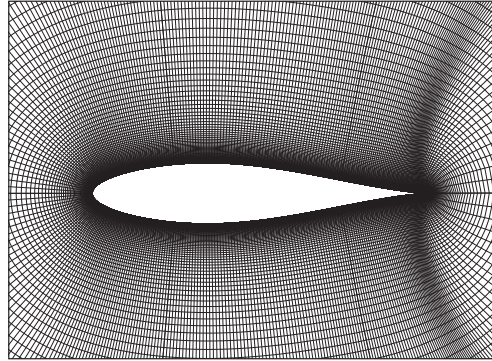
Prediction of airfoil performance at high Reynolds numbers



Test Case, NACA63-018

Computational setup

- ◆ Airfoil computations for $Re = [3, 9, 20]$ million
- ◆ Using three transition models, E^n , $E^n + BP$ and $\gamma - Re_\theta$
- ◆ For natural transition we assume ($N=9$)
- ◆ Mesh resolution 384×256



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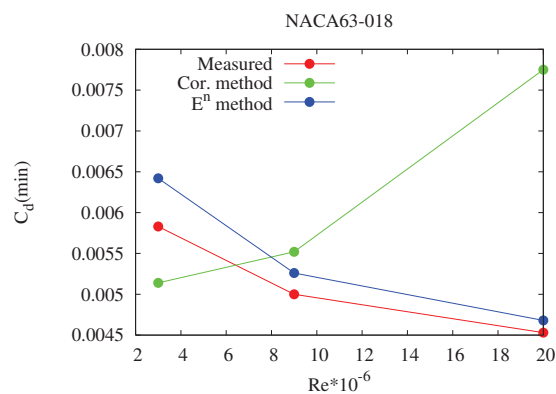
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Prediction of airfoil performance at high Reynolds numbers



Test Case, NACA63-018

Performance for varying Re



The correlation based model do not respond correctly to varying Re !

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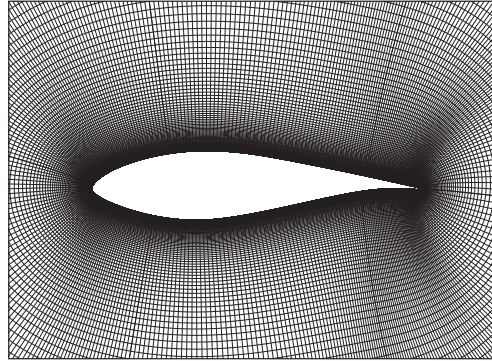
Prediction of airfoil performance at high Reynolds numbers



Test Case, DU00-W-212

Computational set-up

- ◆ Airfoil computations for $Re = [3, 9, 15]$ million
- ◆ Using three transition models, E^n , $E^n + BP$ and $\gamma - Re_\theta$
- ◆ All assuming natural transition ($N=9$)
- ◆ Mesh resolution 384×256



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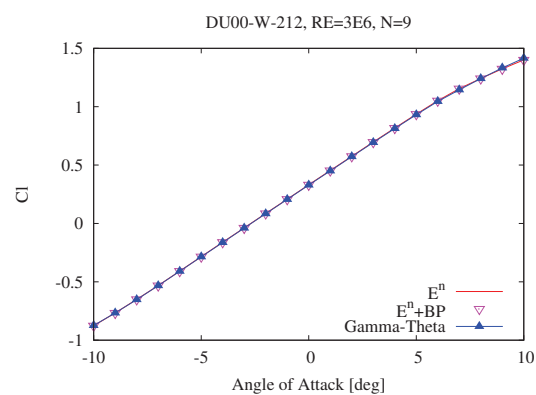
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Prediction of airfoil performance at high Reynolds numbers



Test Case, DU00-W-212

Lift, Natural Transition

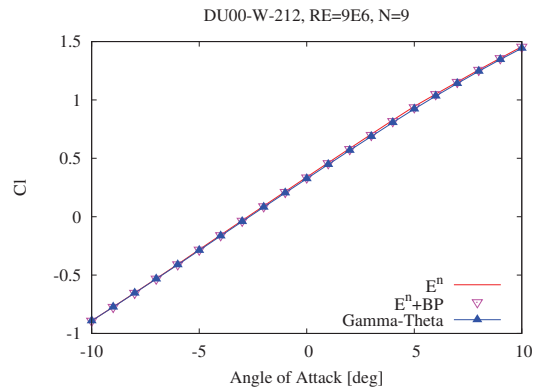


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Niels N. Sørensen, F. Zahle, J.A. Michelsen

Prediction of airfoil performance at high Reynolds numbers

Test Case, DU00-W-212
Lift, Natural Transition

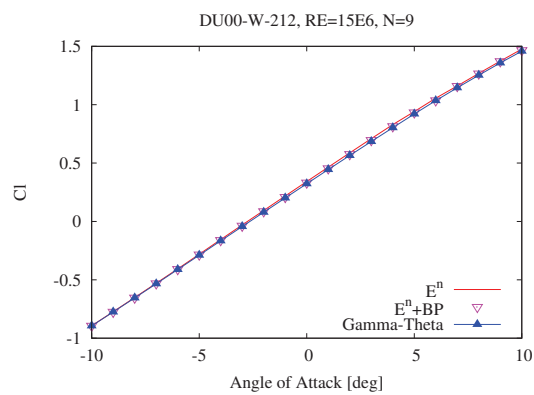


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Prediction of airfoil performance at high Reynolds numbers

Test Case, DU00-W-212
Lift, Natural Transition



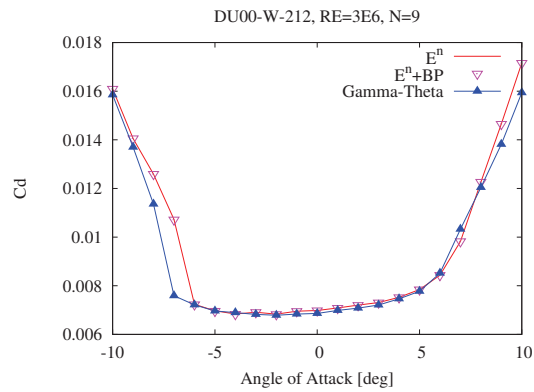
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Niels N. Sørensen, F. Zahle, J.A. Michelsen

Prediction of airfoil performance at high Reynolds numbers



Test Case, DU00-W-212
Drag, Natural Transition



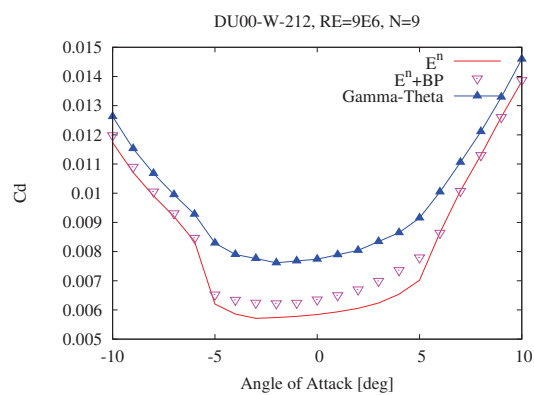
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Prediction of airfoil performance at high Reynolds numbers



Test Case, DU00-W-212
Drag, Natural Transition



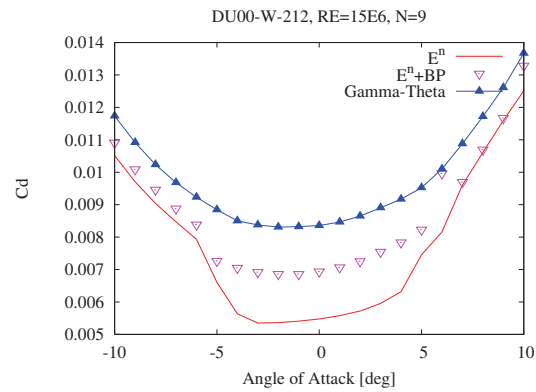
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Niels N. Sørensen, F. Zahle, J.A. Michelsen

Prediction of airfoil performance at high Reynolds numbers



Test Case, DU00-W-212 Drag, Natural Transition



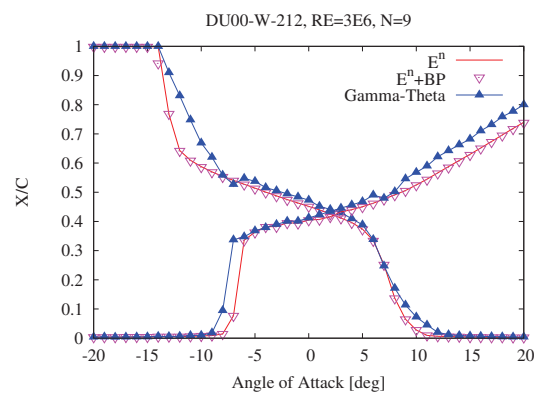
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Prediction of airfoil performance at high Reynolds numbers



Test Case, DU00-W-212 Transition Location, Natural Transition



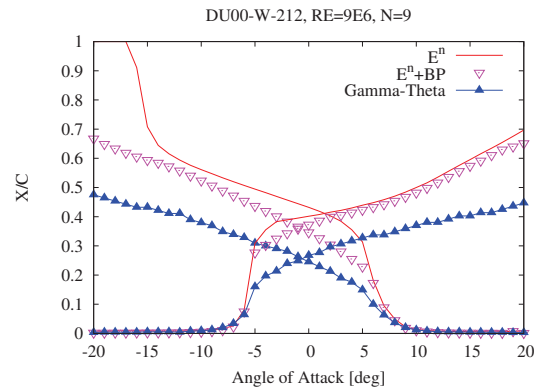
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Prediction of airfoil performance at high Reynolds numbers



Test Case, DU00-W-212
Transition Location, Natural Transition



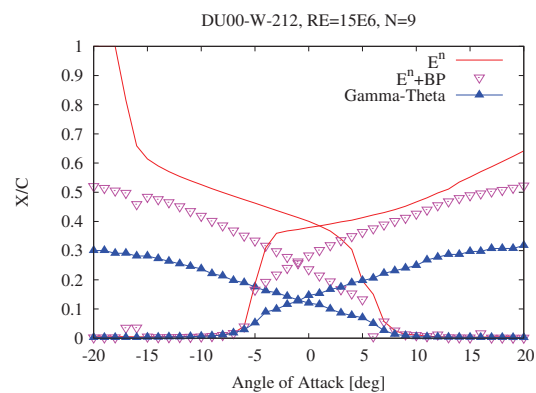
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Niels N. Sørensen, F. Zahle, J.A. Michelsen

Prediction of airfoil performance at high Reynolds numbers



Test Case, DU00-W-212
Transition Location, Natural Transition



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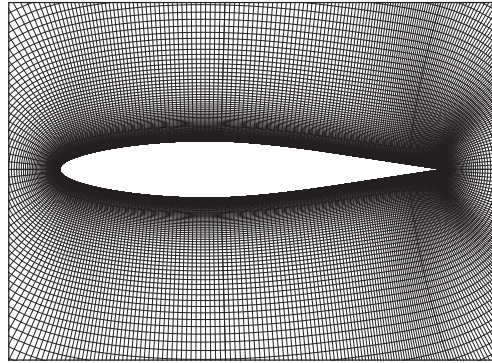
Niels N. Sørensen, F. Zahle, J.A. Michelsen

Prediction of airfoil performance at high Reynolds numbers



Test Case, NACA64₂A015 Computational setup

- ◆ Airfoil computations for $Re=[10:40]$ million, $AOA=0$ deg.
- ◆ Using two transition models, E^n and $\gamma - Re_\theta$
- ◆ Mesh resolution 384×256



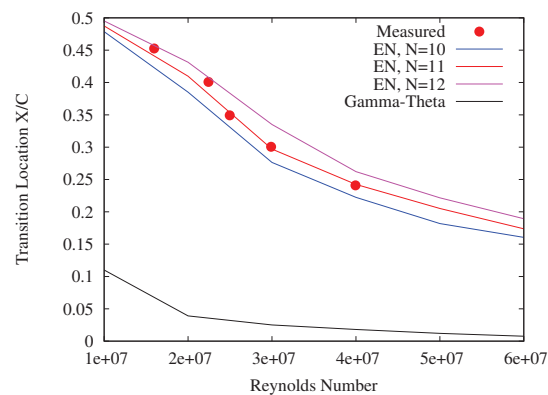
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Prediction of airfoil performance at high Reynolds numbers



Test Case, NACA64₂A015 Performance at high Re



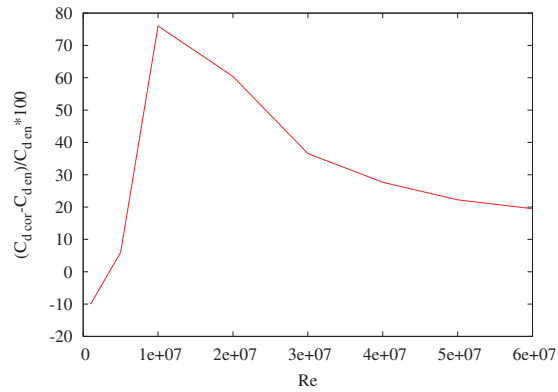
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Prediction of airfoil performance at high Reynolds numbers



Test Case, NACA64₂A015 Performance at high Re



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Prediction of airfoil performance at high Reynolds numbers

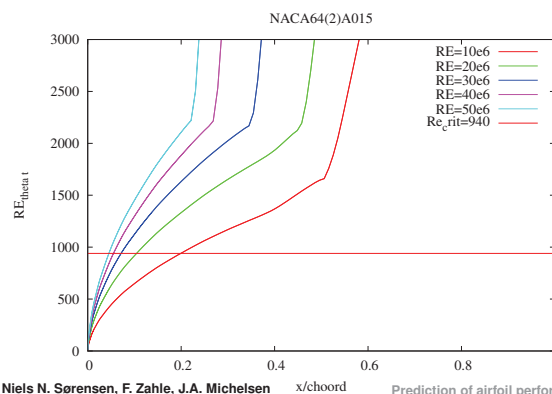


Explanation

Behavior of the correlation based model

The following behavior is observed

- ◆ The Reynolds number is varied through the viscosity
- ◆ The pressure distribution stays nearly constant
- ◆ Turbulent quantities are unchanged away from the airfoil
- ◆ The critical Reynolds number predicted by the $\gamma - Re_\theta$ model stays constant



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Prediction of airfoil performance at high Reynolds numbers



Conclusion

Conclusion and outlook

We need further validation at high Re and relatively low Mach

- ◆ Wind turbine rotors will face high Re with increasing size
- ◆ Lift is weakly dependent on the transition location in normal operation even at high Re
- ◆ The available data show that the $\gamma - Re_\theta$ model over-predict drag at high Re
- ◆ Very little data available for comparison
- ◆ New data will be provided by the AVATAR project for the DU00-W-212 airfoil
- ◆ The National Wind Tunnel could provide data $Re \sim 10$ Mill for low AOA

IEA Task 11/29 MexNext Meeting

NWTC, Louisville, CO, January 11th -13th 2016

CFD simulations of the MexNext rotor experiment using the DLR TAU code

Christoph Heister
Institute of aerodynamics and flow technology
Helicopter Department



Contents

1. Overview of MexNext testcases
2. MexNext CFD simulations using the DLR TAU code
 1. Simulation setup
 2. Mesh
 3. Results
3. Conclusions & Outlook

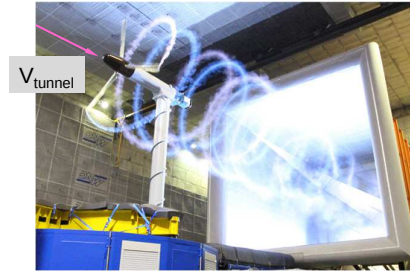


Overview of MexNext testcases

Mexico Rotor in DNW-LLF

D = 4.5m
Pitch = -2.3°
Rpm = 44.52 1/s

Source:
ECN, 2015



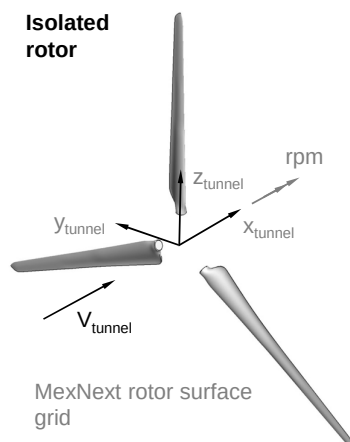
Case	V_{tunnel} (m/s)	ρ (kg/m ³)	T_{inf} (° K)	P_{inf} (N/m ²)
1.1	10.05	1.197	293.63	101398
1.2	15.06	1.191	294.91	101345
1.3	24.05	1.195	294.25	101407

Source: Boorsma, K.; Schepers: J.G.: „Mexnext: Definition of first round of calculations“, October, 2015



MexNext TAU CFD simulations – Simulation setup

Isolated rotor



TAU [1] CFD Setup (RANS, FV):

- Isolated rotor [2], [3]
- Neglection of wind tunnel [4]
- Steady flow assumption
 - Central scheme
 - Backward Euler
 - Scalar dissipation
 - 3v Multigrid
- Fully turbulent flow
 - Wilcox kw 2 eq. turb. model
- 30,000 solver iterations

[1] <http://tau.dlr.de>

[2] Lutz, Th. et. al., 2011

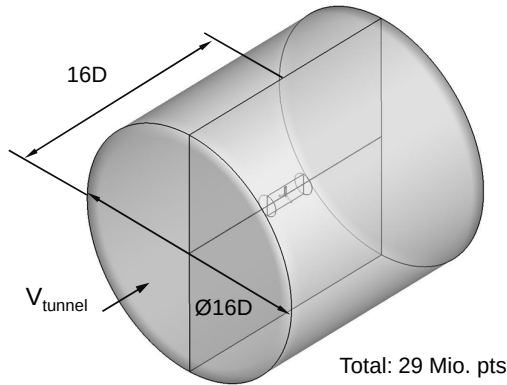
[3] Bechmann, A. et al., 2011

[4] Réthoré, P.-E. et al., 2011

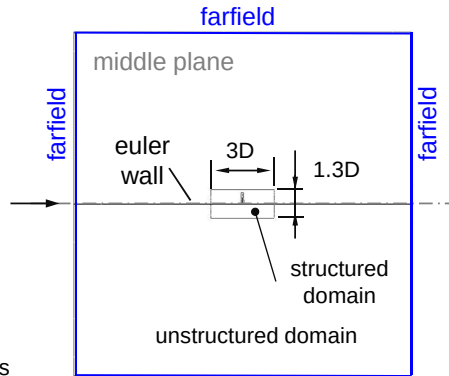


MexNext TAU CFD simulations - Mesh

Grid dimensions:



Boundary conditions:



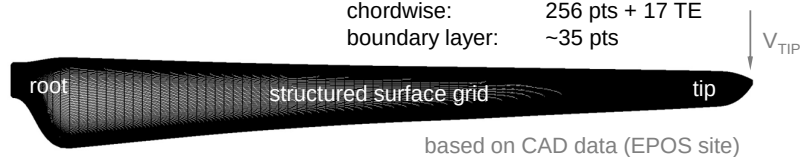
Overset grid (hybrid): Cylindrical background + 3 x blade grids



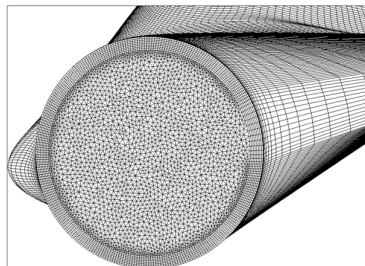
MexNext TAU CFD simulations - Mesh

Blade „Config. 0“ (clean):

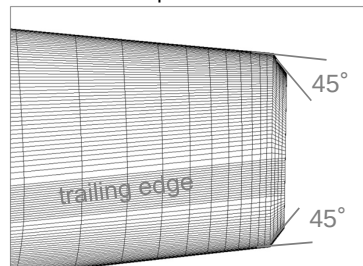
radial: 140 pts
chordwise: 256 pts + 17 TE
boundary layer: ~35 pts



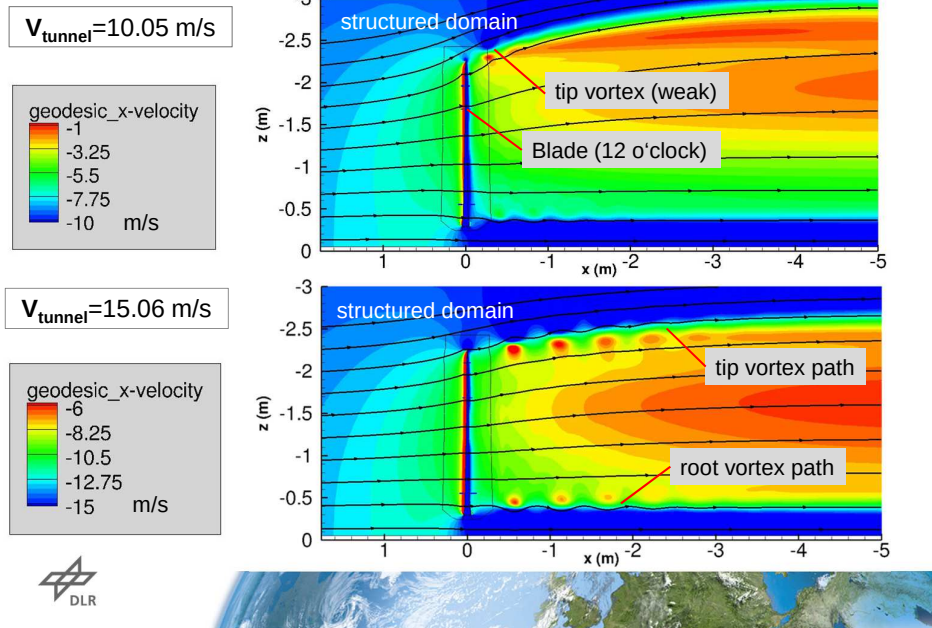
Detail: blade root



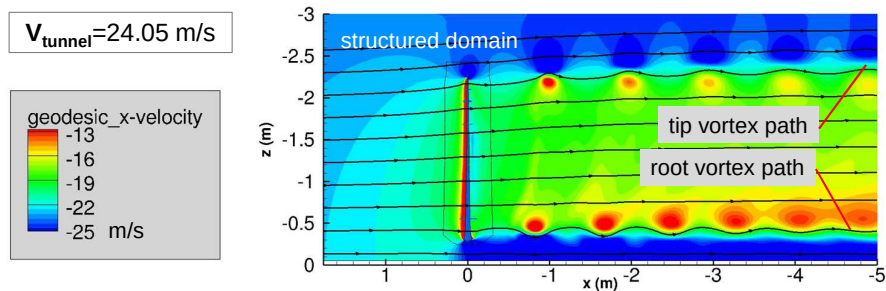
Detail: blade tip



MexNext TAU CFD simulations – Results



MexNext TAU CFD simulations – Results

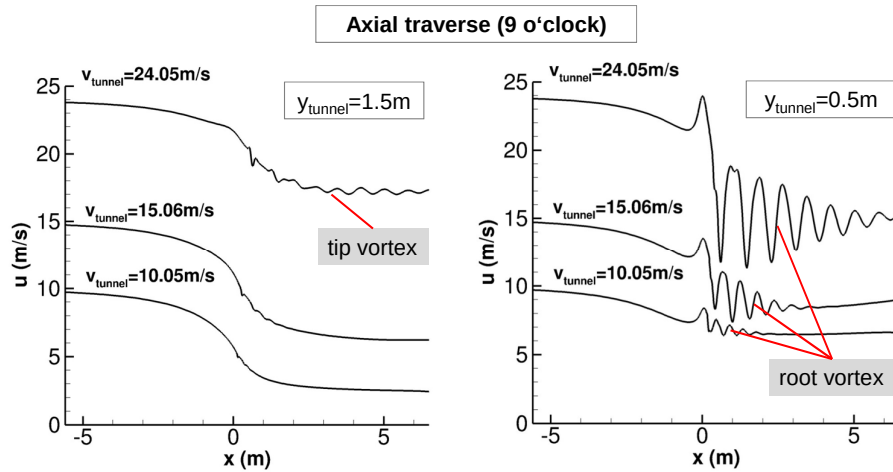


Remarks on TAU-CFD results:

- Wake decelerates and widens as to be expected due to momentum extraction of rotor
- Trajectory of tip vortex can be identified
- Root vortex develops due to neglect of nacelle/spinner
- Vortex intensity and vortex separation increase with V_{tunnel}



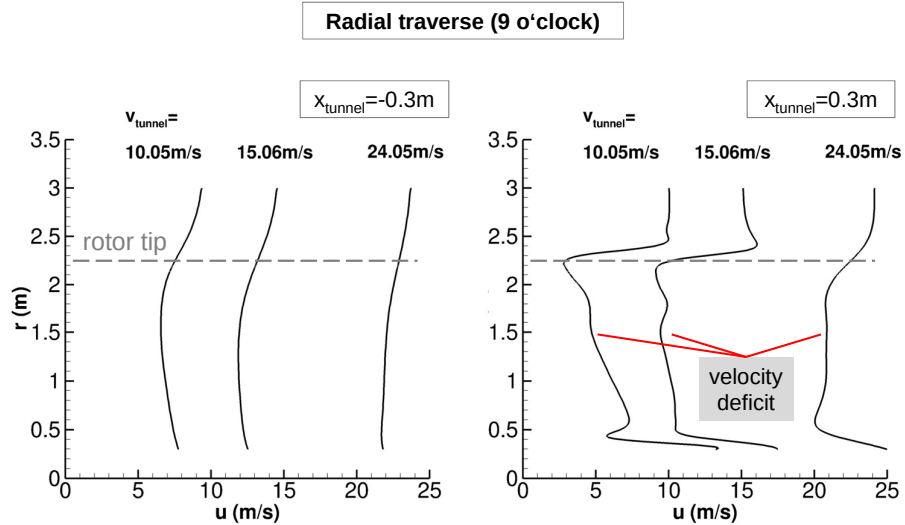
MexNext TAU CFD simulations – Results



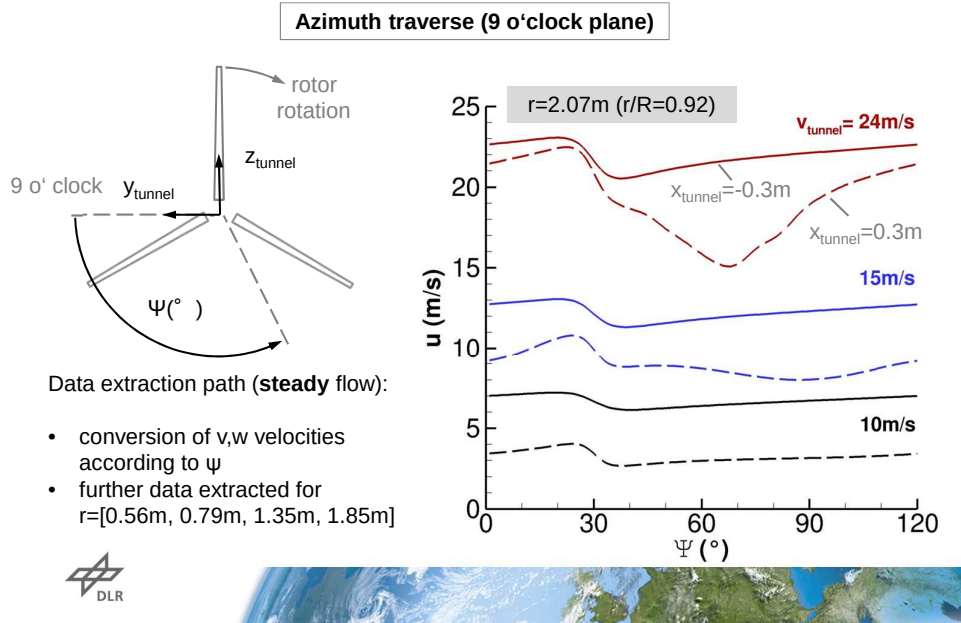
- Outer axial flow: influence of tip vortex visible for $v_{\text{tunnel}} = 24.05\text{ m/s}$
- Inner axial flow: high velocity fluctuations due to root vortex



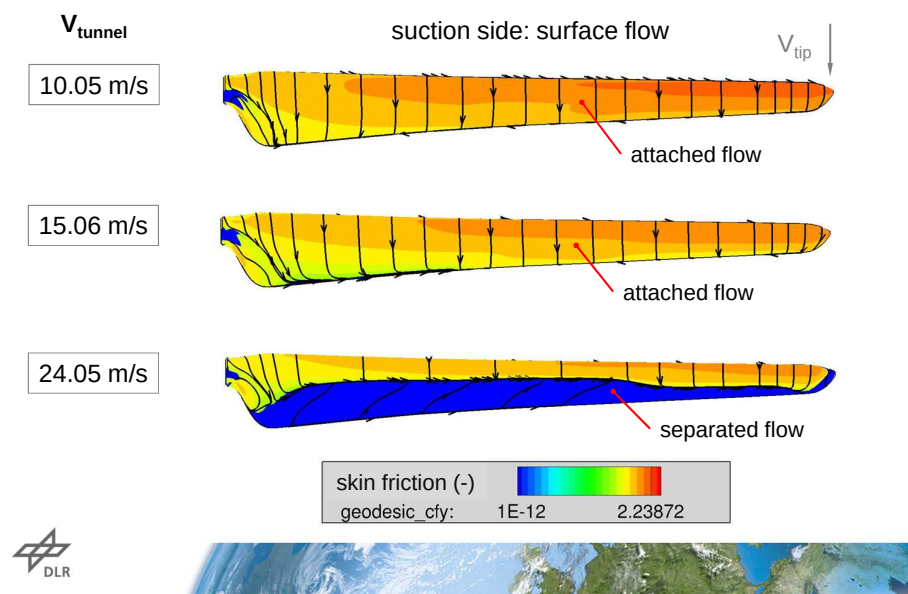
MexNext TAU CFD simulations – Results



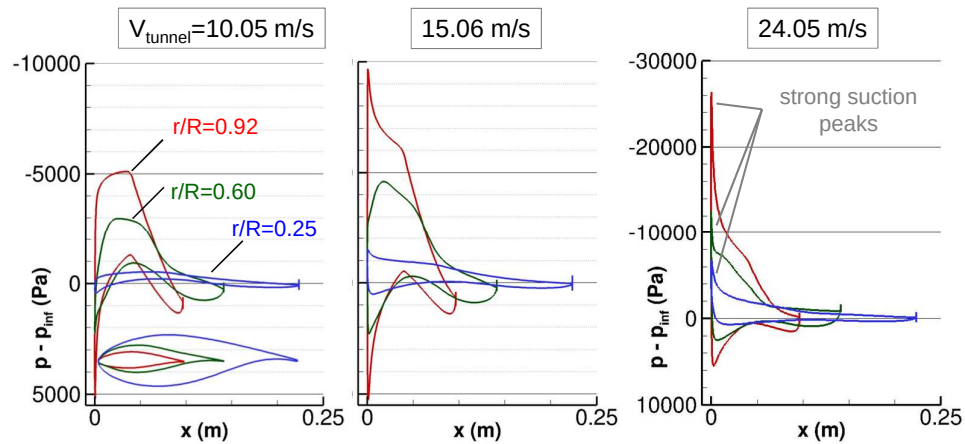
MexNext TAU CFD simulations – Results



MexNext TAU CFD simulations – Results



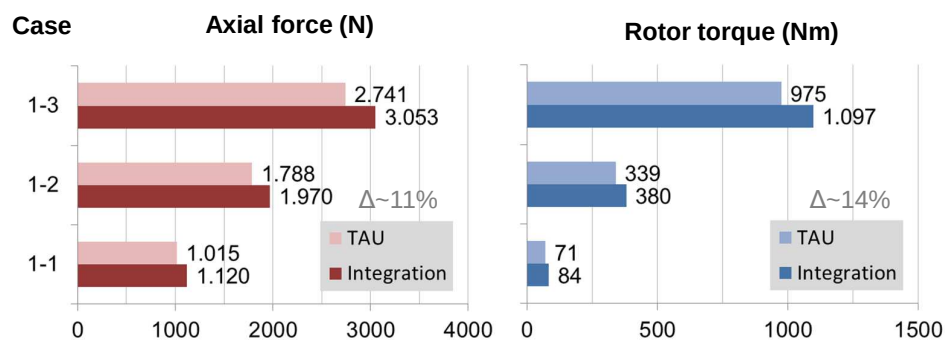
MexNext TAU CFD simulations – Results



Airfoil pressure distributions relative to ambient pressure for three selected rotor blade stations. (not shown for clarity: r/R=0.35, 0.82)



MexNext TAU CFD simulations – Results



Conclusions

- Successful application of DLR TAU code to the MexNext wind turbine testcase (isolated rotor)
- TAU RANS results show physically feasible description of 3D rotor flow field
- TAU results available for improvement of simplified methods

Outlook

- TAU simulations with tripped/free transition
- Unsteady TAU simulation of Case 1.3 (stalled rotor)
- Addition of further wind turbine components (e.g. nacelle/spinner)



Thank you for your attention!



AIAA SciTech 2016, ASME Wind Energy Symposium, San Diego, CA, 4 January, 2016
Modified for IEA 2Tasks 9 & 11 Meeting, NREL NWTC, 11-13 January 2016

Analysis of Nonsteady Transitional Boundary layer Dynamics for Flow over an Oscillating Wind Turbine Airfoil using the γ - Re_θ Transition Model

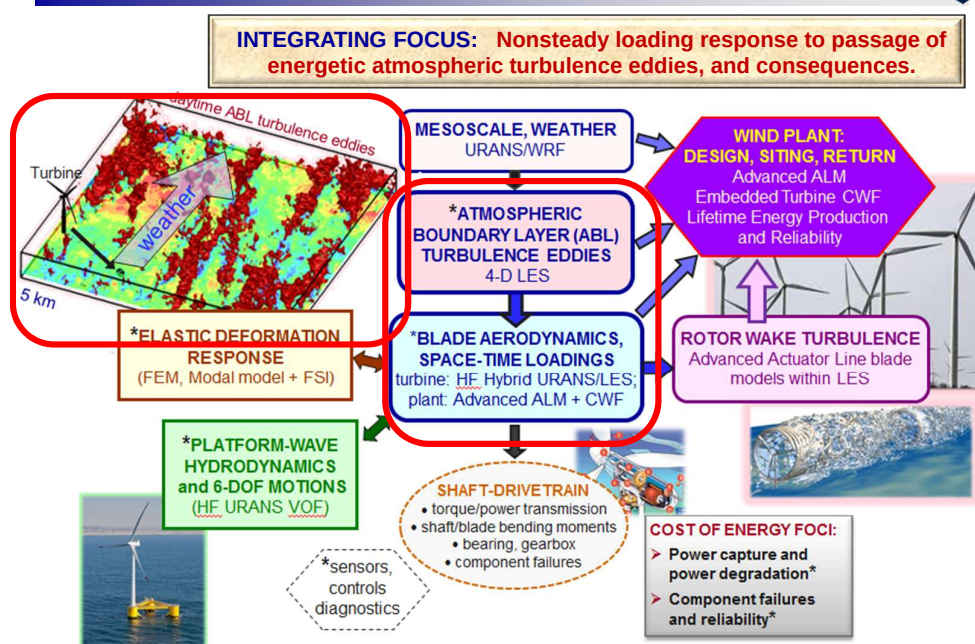
Tarak Nandi, James G. Brasseur*, Ganesh Vijayakumar*
Department of Mechanical and Nuclear Engineering
The Pennsylvania State University
*currently at University of Colorado Boulder
Aerospace Engineering Sciences

FINANCIAL SUPPORT:

- DOE (EERE) Offshore Wind (Technology Development) Program
- PENN STATE: Institutes of Energy and the Environment (PSIEE), CyberScience Institute, Aerospace and Mechanical Engineering

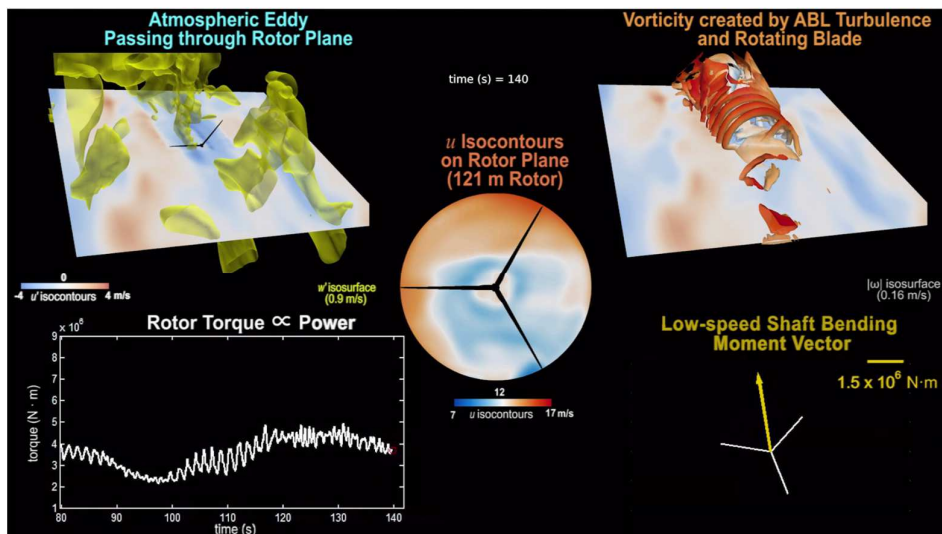


The DOE Penn State “Cyber Wind Facility” Program



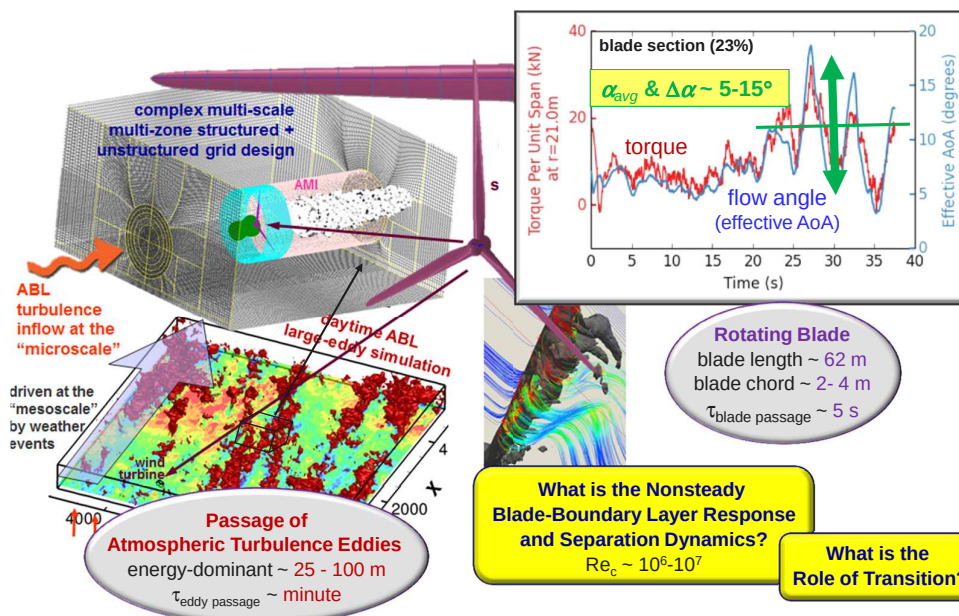
2

Movie of Full Wind Turbine in Atmosphere using ALM: Load and Shaft-Moment Fluctuations with Multiple Atmospheric Eddy Passages



3

Central Role of Nonsteady Changes in Angle of Attack (AoA)



4

Objectives of Current Study



- **Use experiment and simulation of controlled nonsteady changes in angle-of-attack (AoA) to evaluate boundary layer response relevant to wind turbine quasi-periodic nonsteady response to atmospheric eddy passage.**
⇒ oscillating airfoil experiments (S809: Ohio State, Glasgow, Delft)
- **Search for experiment datasets covering a wide nonsteady parameter space covered at Reynolds numbers relevant to transition.**
⇒ Glasgow (Sheng et al., *Wind Energy* 12, 2009)
- **Use computational data generated within the Cyber Wind Facility program (Vijayakumar and Lavelly) to correlate experiment with the nonsteady response of wind turbines operating in the daytime atmosphere.**
- **Replicate with computer simulations using:**
 - advanced URANS TBL model: Menter $k-\omega$ SST model (AIAA J 32, 1994)
 - advanced transition model: Langtry-Menter $\gamma-Re_\theta$ model (AIAA J 47, 2009)
- **Use the above to analyze nonsteady boundary layer response physics, model accuracy, and potential role of transition.**

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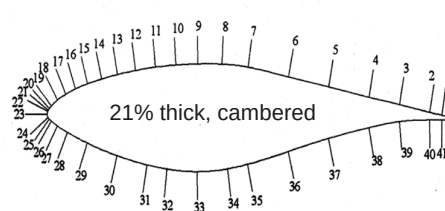
S809 Airfoil Experiments



- **University of Glasgow***
- **Ohio State University (static)**
- **Delft University (static only)**

Glasgow Experiments:

- Freestream Turbulence Intensity:
 $Tu_\infty = 1.5\%$
- **Chord Reynolds number:**
 $Re_c = 1.5 \times 10^6$
- **Static and Oscillating** (except Delft)
- **Clean and "Tripped":**
(60-grit sandpaper strip at leading edge)



Data obtained from Sheng & Coton:

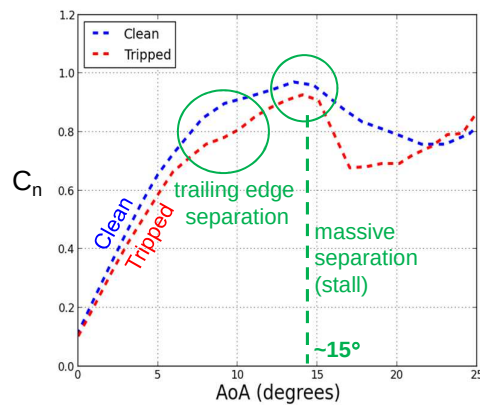
- **Pressure distributions**
- **Force and moment coefficients**
 - integrated from pressure (no shear or drag sensors)

*Sheng Galbraith, Coton, Gilmour, *Wind Energy* 12:752-767, 2009

6

Experiment

Static AoA: Smooth vs. "Tripped"

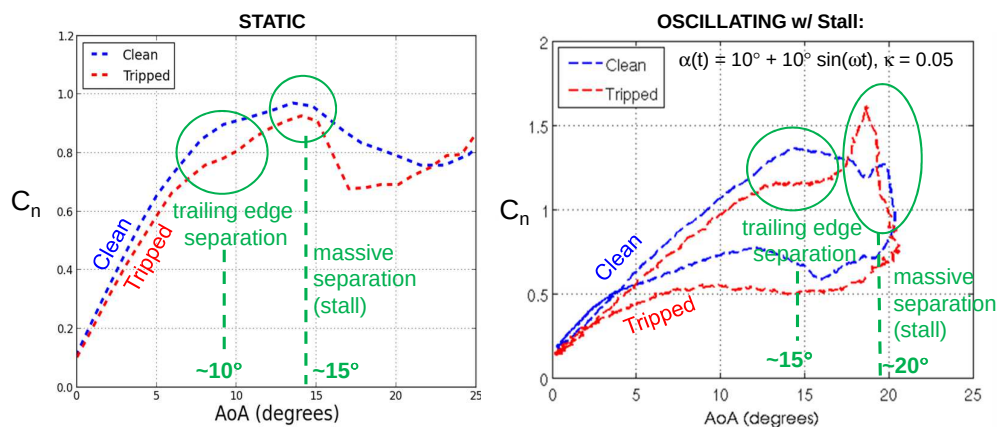


Two phases of boundary layer separation:

- moderate, trailing edge
- massive, leading edge

7

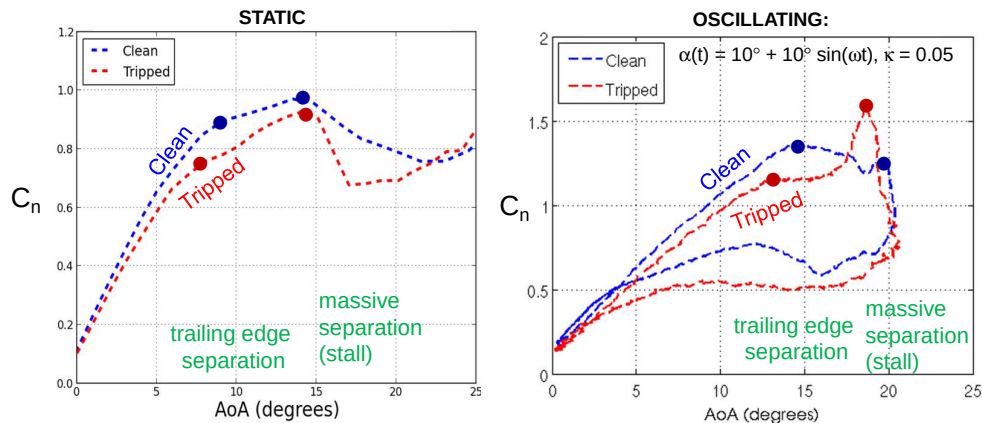
Static vs. Oscillating AoA



- Nonsteady effect $\Rightarrow$ separates at higher AoA
- Nonsteadiness appears as hysteresis

8

The Tripped Boundary Layer vs. the Equilibrium Turbulent Boundary Layer



- The tripped boundary layer separates at lower AoA than does the clean airfoil boundary layer
 - ⇒ The Tripped Boundary Layer is not Fully Turbulent from the Leading Edge (as assumed by RANS models)
 - ⇒ Model predictions can only be fairly compared with clean airfoil data

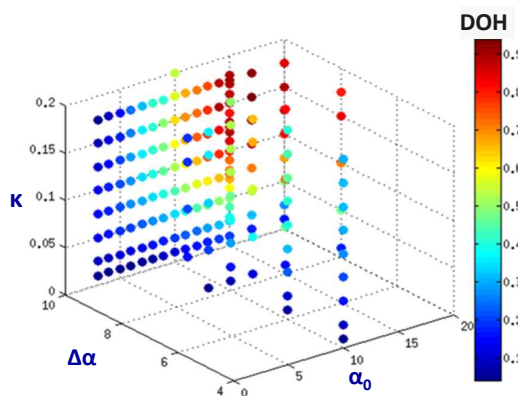
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Nonsteadiness: The Hysteresis Parameter Space

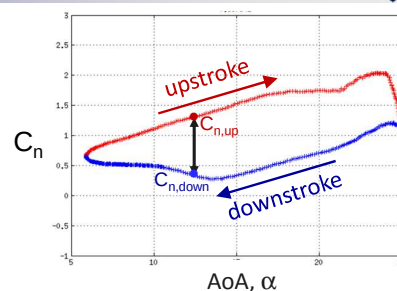


"Degree of Hysteresis" (DOH)

$$\equiv \langle C_{n,up} - C_{n,down} \rangle_{AoA}$$



Hysteresis Parameter Space
from Glasgow data



$$\alpha(t) = \alpha_0 + a_1 \sin(\omega t)$$

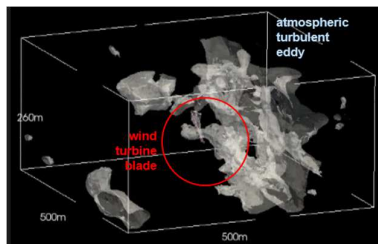
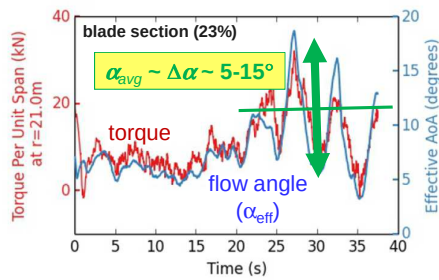
$$\Delta\alpha = \alpha_{\max} - \alpha_{\min} = 2a_1$$

$$\kappa = \frac{\omega}{2U_\infty / c}$$

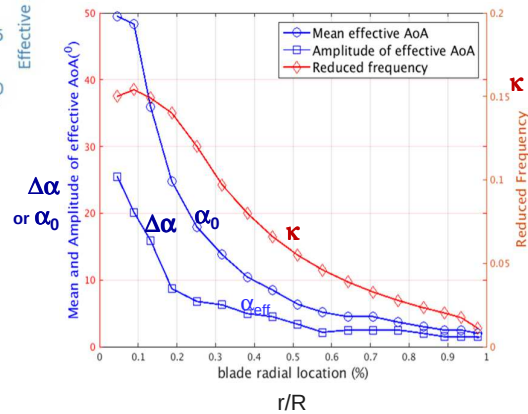
$$\text{DOH} = \text{fun}(\alpha_0, \Delta\alpha, \kappa)$$

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Nonsteady Parameters for Wind Turbines Operating in the Daytime Atmosphere

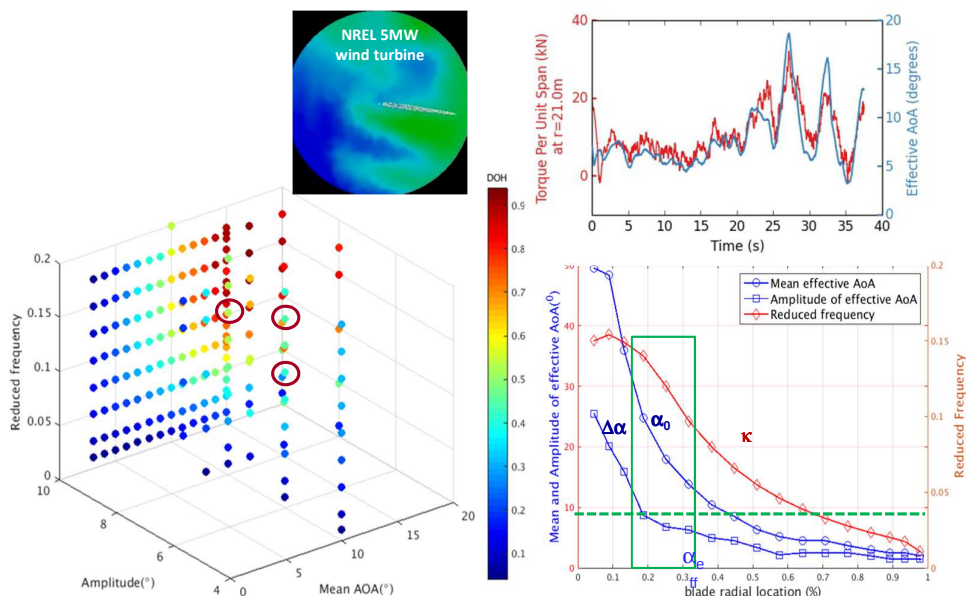


$$\alpha(t) = \alpha_0 + (\Delta\alpha / 2) \sin(\omega t)$$



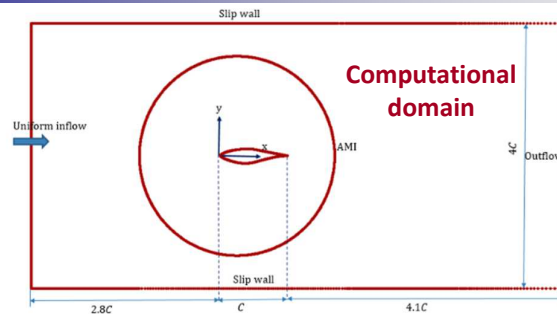
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Wind Turbine Location in the Hysteresis Parameter Space



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Model Predictions and the Role of Transition



- Replicate wind turbine dimensions
- Unstructured Finite Volume solver: OpenFOAM
- SIMPLE-PISO algorithm: semi-implicit, preconditioning
- 2nd order in space and time
- 2 dimensional model
- Arbitrary Mesh Interface (AMI) within OpenFOAM

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The γ - Re_θ Transition Model (brief summary)



Modification of the Menter k - ω SST URANS model

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_j}(\rho u_j k) = \tilde{P}_k - \tilde{D}_k + \frac{\partial}{\partial x_j} \left((\mu + \sigma_k \mu_t) \frac{\partial k}{\partial x_j} \right)$$

$$\frac{\partial}{\partial t}(\rho \omega) + \frac{\partial}{\partial x_j}(\rho u_j \omega) = \alpha \frac{P_k}{v_t} - D_\omega + C d_\omega + \frac{\partial}{\partial x_j} \left((\mu + \sigma_k \mu_t) \frac{\partial \omega}{\partial x_j} \right)$$

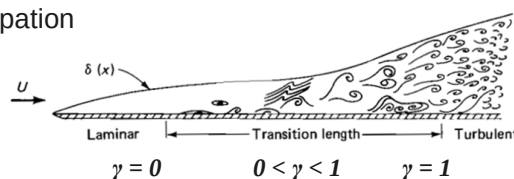
$$\tilde{P}_k = \gamma P_k \quad \tilde{D}_k = \min(\max(\gamma, 0.1), 1.0) D_k$$

"Intermittency" Parameter, $\gamma(x,t)$:

- controls production and dissipation
- controls transition

Central element:

- all variables are local



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Local Transport Model for γ Parameter



$$\frac{\partial(\rho\gamma)}{\partial t} + \frac{\partial(\rho U_j \gamma)}{\partial x_j} = P_\gamma - E_\gamma + \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_f} \right) \frac{\partial \gamma}{\partial x_j} \right]$$

$$P_\gamma = F_{\text{length}} c_{a1} \rho S [\gamma F_{\text{onset}}]^{0.5} (1 - c_{e1} \gamma)$$

where $F_{\text{length}}(\mathbf{x}, t) = \text{fun}\{\overline{\text{Re}}_{\theta t}(\mathbf{x}, t)\}$, $F_{\text{onset}} = \text{fun}\{\text{Re}_v\}$, $\text{Re}_v = y^2 S / \nu$, and

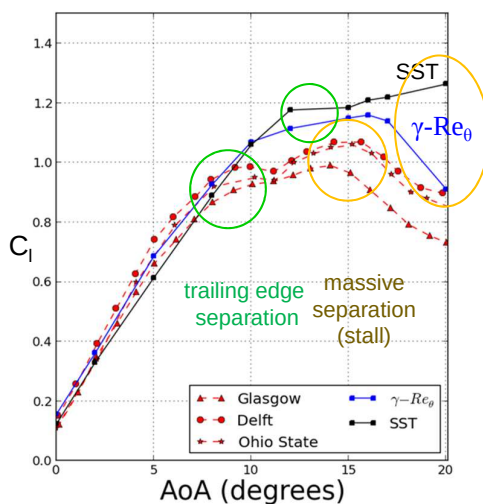
$$\frac{\partial(\rho \overline{\text{Re}}_{\theta t})}{\partial t} + \frac{\partial(\rho U_j \overline{\text{Re}}_{\theta t})}{\partial x_j} = P_{\theta t} + \frac{\partial}{\partial x_j} \left[\sigma_{\theta t} (\mu + \mu_t) \frac{\partial \overline{\text{Re}}_{\theta t}}{\partial x_j} \right]$$

where $P_{\theta t} = c_{\theta t} \rho \frac{\text{Re}_{\theta t} - \overline{\text{Re}}_{\theta t}}{t_u} (1 - F_{\theta t})$ and $\text{Re}_{\theta t}$ is based on an empirical correlation from flat plate data. $F_{\theta t}$ is a blending function

VALIDATION: the transition model was validated against transitional flat plate data
 $\Rightarrow$ errors were found and corrected in publications..

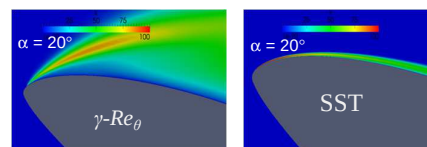
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Comparisons with the Static Clean Airfoil Data



OBSERVATIONS

- Both models predicted trailing edge separation, but delayed to higher AoA
- The transition model captures massive separation, but the SST model does not!



WHY?

- There is a physical basis
- A similar physics occurs with nonsteady hysteresis
- Transition plays a key role!

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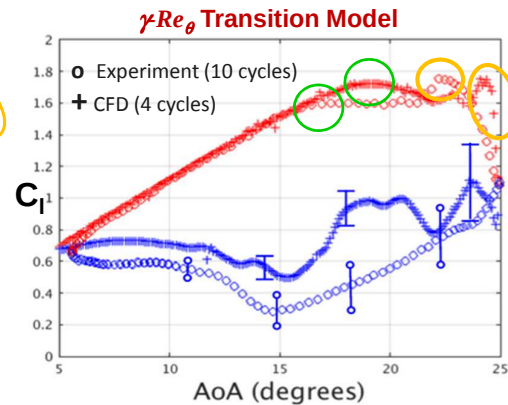
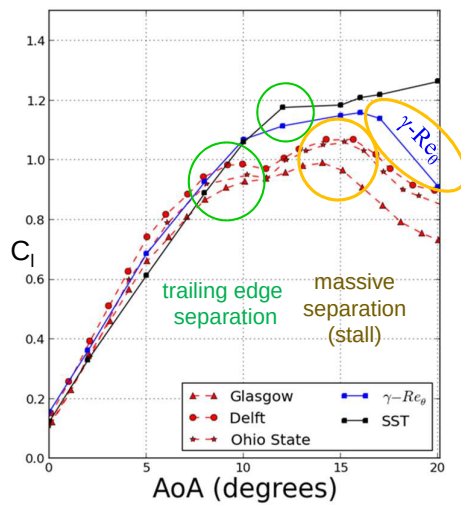
Transition Model

Comparisons with the Static Clean Airfoil Data



A Wind Turbine Case: Deep Dynamic Stall

$$\alpha(t) = 15 + 10 \sin(\omega t), \quad \kappa = 0.1$$



⇒ Delay to higher AoA both in trailing edge separation and in massive stall

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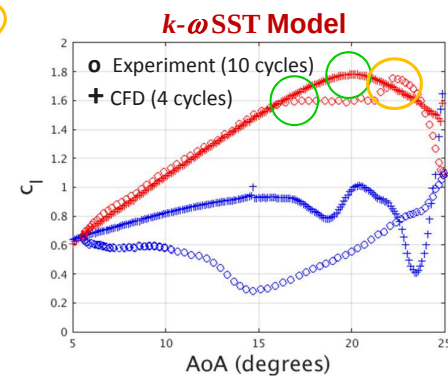
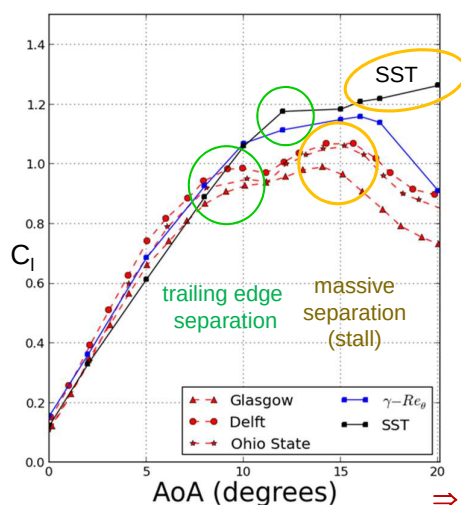
SST Turbulence Model

Comparisons with the Static Clean Airfoil Data



A Wind Turbine Case: Deep Dynamic Stall

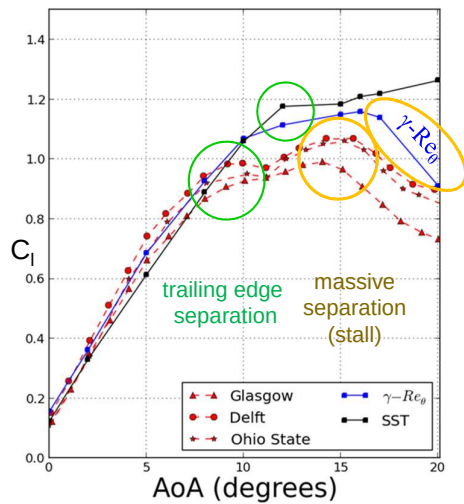
$$\alpha(t) = 15 + 10 \sin(\omega t), \quad \kappa = 0.1$$



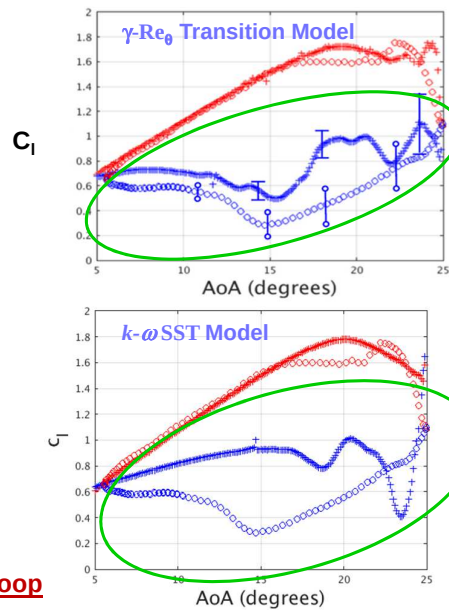
⇒ Delays in trailing edge separation but SST does not capture massive stall!

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The Downstroke

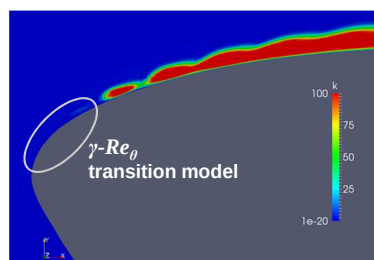


⇒ Delays in TE separation and the inability for SST to capture stall affects the prediction of the return loop

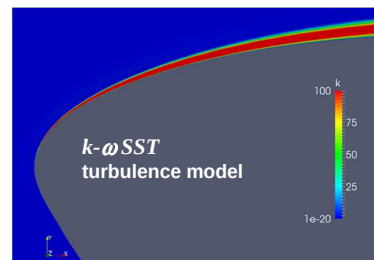


19

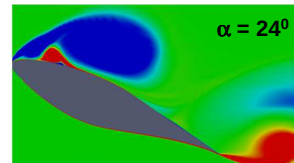
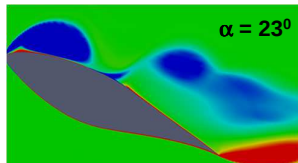
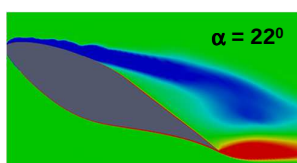
The Central Role of Transition in the Physics of Dynamic Stall and in the Failure of the SST Model



$\alpha = 21^\circ$
upstroke

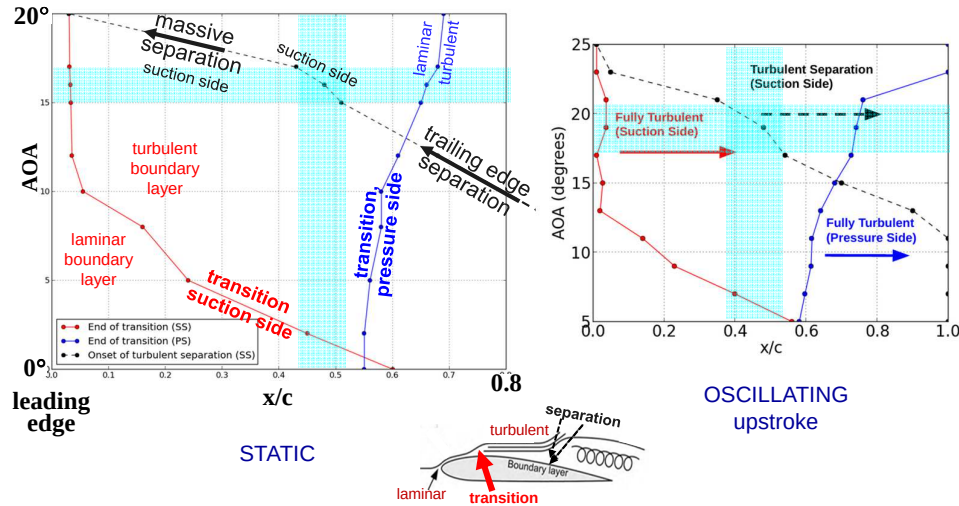


Laminar/transition near the leading edge promotes leading edge vortex formation and massive separation at high AoA that $\gamma-Re_\theta$ captures and SST does not.



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Spatial Relationships between Transition and Separation



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Conclusions



- **Focus on Nonsteady Response of Wind turbine Blades to the Passage of Strong Coherent Atmospheric Turbulence Eddies**
 - Used CFD simulations of turbine blade in atmospheric boundary layer to estimate the three nondimensional parameters that characterize "Degree of Hysteresis"
 - Identified location for wind turbine function within the "Hysteresis Parameter Space" (HPS) of Periodically Oscillating Airfoils
 - Quantified the HPS from a relatively complete experimental dataset of wind tunnel data on the S809 airfoil at $Re_c \approx 1.5 \times 10^6$ by Sheng et al. (2009)
- **Conclusions from Analysis of the Experimental Data**
 - The common claim that reduced frequency < 0.05 is quasi-steady is insufficient
 - The time variations in AoA from the rotation of wind turbine blade through atmospheric eddies creates strong time variations in blade loadings over a large percentage of the blade
 - This conclusion is even more so for sub 1/rev time variations discovered by Vijayalumar et al. (4:30 pm presentation)
 - The "tripped" boundary layer does not behave as a fully turbulent boundary layer as assumed by URANS turbulent boundary layer models.

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Conclusions



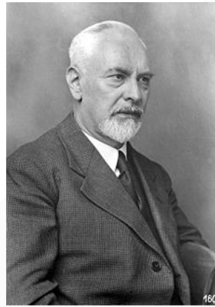
- **Computations of Static and Oscillating Airfoils located at 3 places within the Hysteresis Parameter Space for Wind-Turbine-in-Atmosphere**
 - Compared $k-\omega$ SST turbulent boundary layer model with $\gamma-Re_\theta$ transition model
 - We analyzed static vs. oscillating airfoils for clean airfoil
 - Focused in this presentation primarily on deep stall case
 - **Conclusions from Analysis of the Computations with Experimental Data**
 - Both static and oscillating cases suggest three roughly similar regimes:
 - linear, trailing edge separation, massive separation
 - Both the SST and transition models capture trailing edge separation, but it is delayed to higher AoAs than experimentally measured
 - Massive separation (static) and deep dynamic stall (oscillating airfoils) are sensitive to the transition of the boundary layer from laminar to turbulent near the leading edge
- ⇒ a transition model is required to predict nonsteady behaviors associated with massive separation at high AoAs
- Prediction of delayed trailing edge and massive separation significantly impacts the ability of turbulence models to predict hysteresis accurately, and therefore nonsteady blade loads on wind-turbine blades in atmospheric turbulence.

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Remarks on the Life of Ludwig Prandtl (1875 – 1953) - Slightly corrected since 2014 -



Ca. 1905



Ca. 1937

IEAwind task 29/11 meeting
NWTC, Jan 13, 2016
A.P. Schaffarczyk, UAS Kiel, Germany

Contents

- Introduction
- Prandtl and Wind Turbine Aerodynamics
(The Tip Correction)
- Aerodynamics
- Turbulence
- The Nazi Years (1933 – 1945)
- Nobel-Prize nomination
- Summary

Introduction

3



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Most cited papers (Google Scholar):

Über die Eindringungsfestigkeit (Härte)
plastischer Baustoffe und die Festigkeit
von Schneiden (plasticity, strength)

Über Flüssigkeitsbewegung bei sehr
kleiner Reibung (BL-theory)

4



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[illegible]

85 PhD Students:

Ackeret, Blasius, Betz, Busemann,
Munk, Nirkuasde, Pohlhausen,
Tollmien, Tietjens, von Kármán, and
many others

**Himmelskamp (and Schlichting)
were under supervision of Betz**

Mechanics of Solids:

Plasticity, Solid mechanics

Not mentioned here

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Sources

1. Scientific work

- Vier Abhandlungen (4 Selected Papers)
 - Darrigol: World of FLOW
 - **Eckert: The Dawn of Fluid Dynamics**
- **Discipline between Science and Technology** -
→ **Biography to appear 2016**
 - Anderson: A History of Aerodynamics
 - Rotta: Die Aerodynamische Versuchsanstalt in Göttingen
 - Wuest: Sie zähmten den Sturm (Taming the tempest)
 - Bodenschatz and Eckert: Prandtl and the Göttingen school
- In: Davidson et al (Eds.): A Voyage Through Turbulence

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2. Person and Politics:

J. Vogel-Prandtl: Ludwig Prandtl – Ein Lebensbild
(available in English from ICTP)

v. Kármán/l. Edson: The Wind and Beyond

Trischler: Luft- und Raumfahrtforschung in
Deutschland 1900 – 1970

Maier (Hrsg.) Rüstungsforschung im
Nationalsozialismus

C. Tollmien(!) KWI und AVA in: Die Universität
Göttingen unter dem Nationalsozialismus

Vier Abhandlungen (1925)

1. Boundary Layer Theory
2. Lifting Line Theory, Part I
3. Lifting Line Theory, Part II
4. Screw Propellers with lowest
resistance (By A. Betz with an
Appendix by L. Prandtl)

Wind Turbine Aerodynamics

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Paper 4 relates most to Wind Turbine Aerodynamics,
„Zusatz“ (Appendix) by Prandtl of
Betz' Screw propellers with lowest resistance

90 A. Betz.

$$\alpha = \frac{H}{n} \cdot \cos \varepsilon = \frac{2\pi r B}{n \sqrt{v^2 + H^2 u^2}} \quad (4)$$

An die Stelle der Umströmung der Ränder der Schraubenflächen mag nun die Umströmung der Ränder eines äquivalenten Ebenensystems von Abstand α , gemäß Fig. 5, gesetzt werden, um den Abfall der Zirkulation an den Flügelspitzen zu studieren. Nach den Darlegungen von Nr. 6 der II. Mitteilung ist die Zirkulation gleich dem Potentialsprung der in der Flüssigkeit zurückgelassenen Strömung zu setzen. Es ist also das Strömungspotential für die in Fig. 2 angedeutete Strömung zu berechnen. Die nach Fig. 2 durchschnittene Ebene (x -Ebene) läßt sich knüpfen auf das Äußere des Einheitskreises (x -Ebene) abbilden durch

$$x = \frac{a}{\pi} \ln \frac{1}{2} \left(t + \frac{1}{t} \right) \quad (5)$$

Die Strömung von Fig. 2 verwandelt sich dabei in zirkulatorisches Umströmen des Einheitskreises, es ist also $\Phi + i\Psi = F - i\pi\alpha x$, hiernach ergibt sich nach kurzer Rechnung:

$$x = \frac{a}{\pi} \ln \cos \frac{F}{\pi} \quad (6)$$

Für reelle Werte von F , also für die Stromlinien $\Psi = 0$, wird

$$x = \frac{a}{\pi} \ln \cos \frac{\Phi}{\pi} \quad \text{oder} \quad \Phi = \pm \pi \arccos e^{\pm \pi x/a}$$

für $x \geq 0$. Der Potentialsprung wird somit, wenn, mit $R = \text{Äußerradius}$, $R - r$ für $-x$ gesetzt wird,

$$[\Phi] = 2\pi \arccos e^{-\pi(R-r)/a} \quad (7)$$

Der Abfall der Zirkulation an den Flügelspitzen der Schraube wird nun proportional dem Abfall des Potentialsprungs in Gleichung (7) ausgedrückt. Bei den praktisch üblichen Annahmen liegen die Funktionswerte von Γ nach Gleichung (3) für die Gegend der Flügelspitzen regelmäßig schon in der Nähe des Asymptotenwertes, sind also dort fast konstant. Es wird deshalb kein großer Fehler gegenüber der wahren Lösung der Minimaufgabe entstehen, wenn man an ihrer Stelle denjenigen Verlauf von Γ setzt, der sich ergibt, wenn man die Werte von Γ aus Gleichung (3)

Schraubenpropeller mit geringstem Energieverlust. 91

mit dem Ausdruck der Gleichung (7) multipliziert, wobei durch die Wahl $\varepsilon = 1/2$ der Asymptotenwert dieses Ausdruckes auf 1 normiert wird. Hiernach ergibt sich also als Endformel:

$$\Gamma = \frac{4u}{\pi} \cdot \frac{(v+u/2)r/a}{(v+u/2)^2 + r^2/a^2} \cdot \arccos e^{-\pi(R-r)/a} \quad (8)$$

mit α nach Gleichung (4), in der ebenfalls gemäß der Anmerkung S. 60 v durch $v+u/2$ ersetzt werden kann. Die Gleichung gilt um so genauer, je mehr Schraubenflügel vorhanden sind, es ist aber zu erwarten, daß sie auch bei wenigen, ja selbst bei zwei Flügeln, immer noch brauchbare Resultate ergibt. Die Fig. 3 zeigt in der stark vergrößerten Linie den Verlauf von Γ abhängig von r für vier Flügel und für einen Wert

$$H/2\pi R = (v+u/2)/R\alpha = 1:5,$$

was etwa dem bei Flugerschrauben Üblichen entspricht. Der Verlauf von Γ bei zwei Flügeln ist zum Vergleich in Fig. 3 gestrichelt beigefügt.

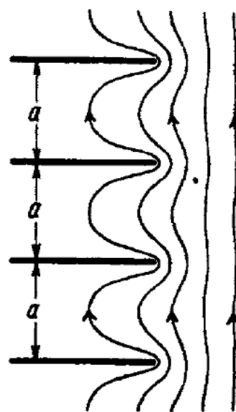
Bei Vergrößerung der Flügelzahl würde die Größe α verkleinert, der Abfall an der Flügelspitze also steiler werden; im Grenzfall von unendlich vielen Flügeln wäre dieser Abfall also senkrecht anzunehmen. Der Verlauf an Volligkeit in der Verteilung Γ und daher auch in der Schubverteilung, der bei endlicher Flügelzahl gegenüber unendlich vielen Flügeln eintritt, läßt sich dadurch zum Ausdruck bringen, daß man die Verteilung nach Gleichung (7) gleichzeitig durch ein Rechteck ausgleicht. Dadurch erweist sich irgend eine Schraube als dynamisch gleichwertig mit einer Schraube mit unendlich vielen Flügeln, bei der die Flügel um einen Betrag a' (Fig. 3) kürzer sind als die Flügel der gegebenen Schraube. Der Betrag a' ergibt sich aus der Forderung, daß

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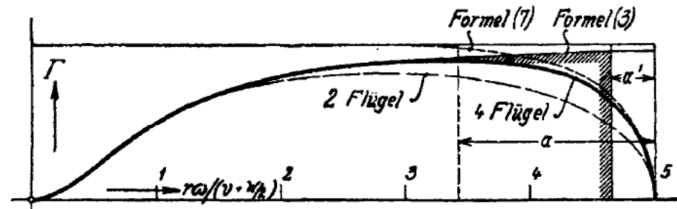


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Figur 2.



Figur 3.

Formel: equation
Flügel: blade

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Shen et al, 2005
Tip loss correction for wind
turbine computations

E. Branlard, 2011/2013
Wind turbine tip-loss correction

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$$F_{C_n, Sh} = \frac{2}{\pi} \arccos \exp \left[-g(\lambda) \frac{B(R-r)}{2R \sin \phi_R} \right]$$

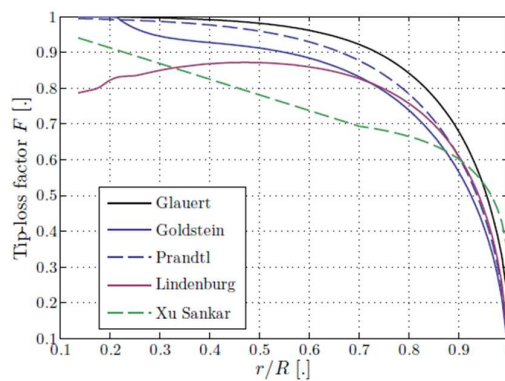
$$g(\lambda) = \exp[-c_1(B\lambda - c_2)] + c_3$$

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From Shen et al (2005)



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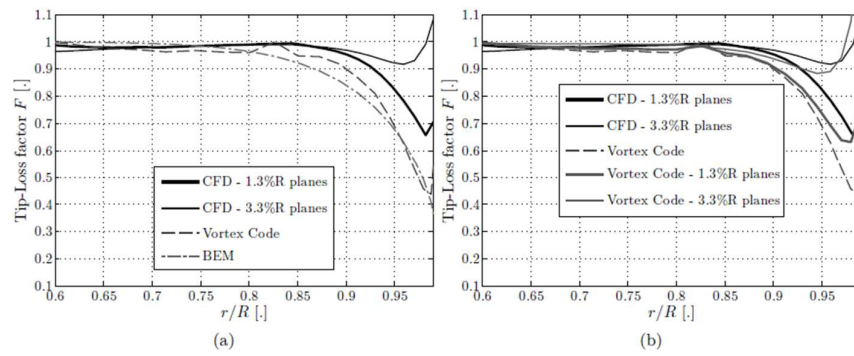
From Branlard (2013)

16



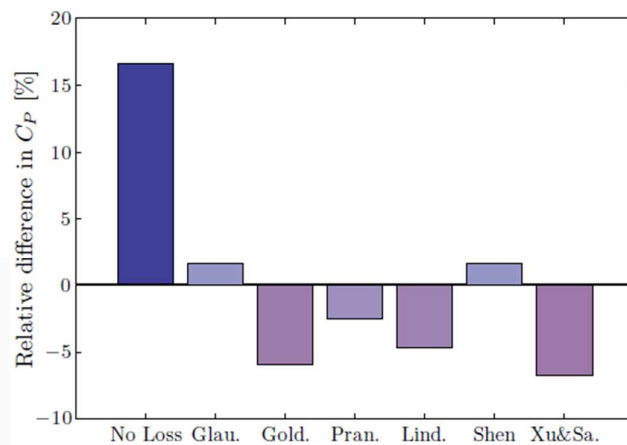
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From Branlard (2013)

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Another 5% rule

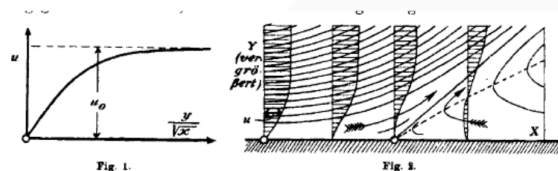
Aerodynamics

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Paper 1

Boundary Layer Theory (1904)

Overcoming d'Alembert's Paradoxon: Viscous Drag



Singular Perturbation Theory
Matched asymptotic Expansion
CFD (laminar flow)

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Paper 2 and 3:

Lifting Line Theory I (12.1918) and II (02.1919)
Overcoming d'Alembert's Paradoxon: Lift

Kutta-Joukowski-Theorem

$$L = \rho V \Gamma$$

2D ideal flow, easy to prove, but

For Fixing Circulation Γ an empirical Kutta Condition has to be used

This is very delicate (no infinite velocities ...)

Is there a „derivation“ from BL theory ?

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Results:

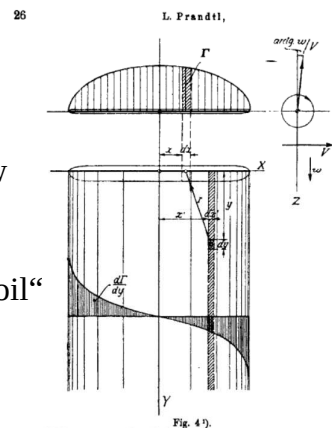
- For finite wings lift “induces” drag
- Induced drag is minimized when circulation is elliptically distributed

- Much more easier to see from H. Glauert's

The elements of aerofoil and airscrew Theory (1926) (10.31, page 134)

See Bloor's „The Enigma of the Airfoil“

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Back to paper 4

A. Betz' PhD thesis (03.1919)

Applied paper 2 and 3 to a (ship) Propeller

Die Zirkulation für einen Schraubenflügel im Radius r ergibt sich bei n Schraubenflügeln, die gleichmäßig auf den Umfang verteilt sein mögen, zu

$$\Gamma = \frac{2\pi r w_t}{n} = \frac{2\pi w}{n} \cdot \frac{v r^2 \omega}{v^2 + r^2 \omega^2} \quad (= \frac{H \cdot w_n}{n}), \quad (3)$$

23

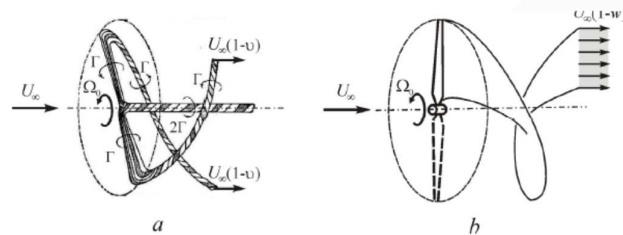


Figure 4.1. Sketch of the vortex system corresponding to lifting line theory of the ideal propeller of Joukowski (a) and Betz (b).

From: Okulov, Soerensen and van Kuik:
Development of the optimum rotor theories

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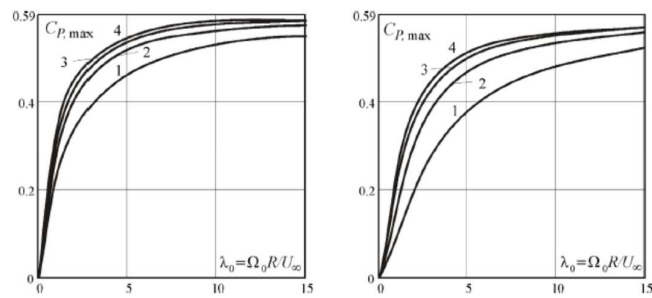


Figure 4.4. Wind power coefficients, C_p , of an optimum rotor as function of tip speed ratio and number of blades. Left: Joukowsky rotor; Right: Betz rotor.

From: Okulov, Soerensen and van Kuik: Development of the optimum rotor theories

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Turbulence

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Turbulence

1. Mathematical definition

We ask for a proof of one of the following four statements (Fefferman, 2000):

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(A) Existence and smoothness of Navier–Stokes solutions on $\mathbb{R}^3$. Take $\nu > 0$ and $n = 3$. Let $u^\circ(x)$ be any smooth, divergence-free vector field satisfying (4). Take $f(x, t)$ to be identically zero. Then there exist smooth functions $p(x, t)$, $u_i(x, t)$ on $\mathbb{R}^3 \times [0, \infty)$ that satisfy (1), (2), (3), (6), (7).

(B) Existence and smoothness of Navier–Stokes solutions in $\mathbb{R}^3/\mathbb{Z}^3$. Take $\nu > 0$ and $n = 3$. Let $u^\circ(x)$ be any smooth, divergence-free vector field satisfying (8); we take $f(x, t)$ to be identically zero. Then there exist smooth functions $p(x, t)$, $u_i(x, t)$ on $\mathbb{R}^3 \times [0, \infty)$ that satisfy (1), (2), (3), (10), (11).

(C) Breakdown of Navier–Stokes solutions on $\mathbb{R}^3$. Take $\nu > 0$ and $n = 3$. Then there exist a smooth, divergence-free vector field $u^\circ(x)$ on $\mathbb{R}^3$ and a smooth $f(x, t)$ on $\mathbb{R}^3 \times [0, \infty)$, satisfying (4), (5), for which there exist no solutions (p, u) of (1), (2), (3), (6), (7) on $\mathbb{R}^3 \times [0, \infty)$.

(D) Breakdown of Navier–Stokes Solutions on $\mathbb{R}^3/\mathbb{Z}^3$. Take $\nu > 0$ and $n = 3$. Then there exist a smooth, divergence-free vector field $u^\circ(x)$ on $\mathbb{R}^3$ and a smooth $f(x, t)$ on $\mathbb{R}^3 \times [0, \infty)$, satisfying (8), (9), for which there exist no solutions (p, u) of (1), (2), (3), (10), (11) on $\mathbb{R}^3 \times [0, \infty)$.



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Otelbaev (02.2014) I have proven that ! (definitely wrong)

Tao: „finite time blow-up“ may be possible for NSE

Birniir: Millenium Problem may cover only laminar(!)
„smooth“ solutions

→ **more in my 2017 talk**

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2. From Physics:

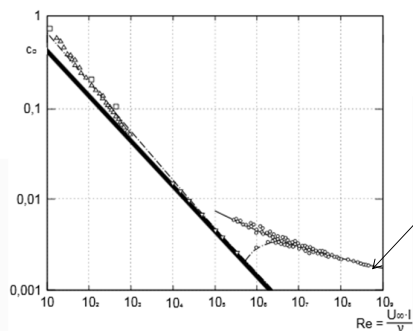
Sommerfeld, Heisenberg, Onsager, Kolmogorov, Kraichnan and many others tried to find a „solution“. But with limited success only.

Prandtl's mixing length approach is an adaption from kinetic theory of gases to Turbulenzballen (lumps of turbulence)

Successful to jets and still in use for LES

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3. Engineering approach: Ask not „why“ but „how“ !



Example
Turbulent Flat plate drag
Prandtl's mixing length
approach gives

$$\frac{1}{\sqrt{\lambda}} = 2.0 \log(Re_D \sqrt{\lambda}) - 0.8.$$

(D = diameter) Pipe flow

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Less known:

Prandtl (and Kolmogorov) published both in 1945
one (two)-differential-equation models of turbulence

$$\nu_t = k^{\frac{1}{2}} l = C_D \frac{k^2}{\varepsilon}$$

$$\frac{\partial k}{\partial t} + U_j \frac{\partial k}{\partial x_j} = \tau_{ij} \frac{\partial U_i}{\partial x_j} - C_D \frac{k^{\frac{3}{2}}}{l} + \frac{\partial}{\partial x_j} \left[\left(\nu + \frac{\nu_T}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right]$$

Solved by Wiegardt for a turbulence grid and 2D
channel flow

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1933 – 1945
The Nazi Years

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The Nazi years (1933 – 1945)

- Jan. 1933: Hitler elected
- March 1933: Ermächtigungsgesetz (enabling act)
- April 1933: law to fire non-Arian (Jewish) civil servants
- April 1936: „Vierjahresplan“ (Armament plan)
- Nov. 1938: Reichspogromnacht („Crystal Night“)
- Sept. 1939: WW2 started by attacking Poland
- June 1940: France attacked
- July 1941: Soviet Union attacked
- Jan. 1942: Wannseekonferenz (organizing Holocaust of European Jews)
- Feb. 1943: Battle of Stalingrad – the turning point
- 8. Mai 1945: Germany surrenders

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What did the change mean to Prandtl ?

- A lot of able colleagues were fired (Sometimes he tried to help them)
- Lots of governmental funding, but
- AVA was much smaller (less than 700 empl.) than DVL (2000 empl.)
- Germany's air force policy then was (sometimes) a technical success but may be regarded as a failure in general
- Produced airplanes at war-times:
Germany 100 k, GB + USA 400 k

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The 1938 Congress in USA:

From v. Kármán/Edson (1967):

„You should not believe only the bad things about the Hitler regime”, Prandtl replied. “Maybe there are times in history when order and discipline are more important than justice and freedom”

More along the same line from other German documents

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Compare to:
W. Heisenberg
W. von Braun

German Atomic Bomb Project
V2 and US (Moon) Rockets



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D. Cassidys Biography „Uncertainty“

explains the behaviour of Heisenberg very well.

Nobody knows what he really thought about the German bomb.

After the war he was very much isolated internationally.

See:

M. Walker, Die Uranmaschine (Uranium Engine)

M. Frayn, Kopenhagen

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von Braun was member of NSDAP (Nazi-Party) and SS.

At „Schacht Dora“ (Production site of V2) about 20 000 war-prisoners were killed.

About 6000 rockets were manufactured; 3200 of them fired and killed additional 8000 civilian people.

However he became important to the US.



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Now: How much a Nazi was Prantl ?

*„Dem Staat und der Wissenschaft dienen“
(Serve the state and science)*

„Entnazifizierungs“ (de-nazification) grades from about
3M (western Allied) investigations:

- | | |
|---------------------------|----------------------|
| 1. War Criminals | ~ 100 far too much ! |
| 2. „Belastete“ (charged) | ~ 30 k |
| 3. Less charged | ? |
| 4. „Mitläufer“ (follower) | 50 % |
| 5. Investigation closed | 35 % |

His own statement (1946): „Mitläufer“ (follower)



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Nobel-Prize nomination

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Q: What Nobel Prize Winners have in common ?

A: (E. Fermi): Nothing – not even intelligence

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Nobel Awards of the 20th century
Mainly dominated by
Quantum Physics:
Atoms and Nuclei, Particles & Fields

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J.D. Anderson, Physics Today (2005):

„By every right, his boundary-layer concept was worthy of the Nobel Prize. He never received it, however; some say the Nobel Committee was reluctant to award the prize for accomplishments in classical physics”

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C.W. Oseen (1879 – 1944, Swedish theoretical physicist) states in 1928:

The propositioners apparently assume that in order to receive a physics Nobel Prize, a long-standing, dynamic and successful activity within physics, including mechanics suffices. If this assumption were in agreement with the Charter of the Nobel foundation, I would have recommended that professor Prandtl be awarded a physics Nobel prize. However, since a prerequisite for receiving such a prize according to the Charter, is a **discovery or invention in the field of physics**, and since no discovery or invention of the magnitude that would justify a Nobel Prize has been put forward by the propositioners, I conclude that I cannot second the nomination

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Apparently Bragg (Jn.) nominated Prandtl in 1935 for his work on turbulence.

Prandtl replied:

„It is extremely friendly of you to be of the opinion that i am a suitable candidate for the Nobel Prize. However, i have no hopes in this direction because I cannot consider the work really part of physics. . .“

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Remark: Arnold Sommerfeld
Was nominated 81 times (!)

- Orr-Sommerfeld Stability Equation
- Lubrication Theory
- Supervision on Heisenberg's (failed) PhD thesis on Turbulence



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Physics or Engineering ?

1909: wireless telegraphy
1912: Lighthouse
2014: blue LEDs

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Summary

- Outstanding Contributions to Aerodynamics
- Close but 2nd to Nobel-Prize level
- Benefitted from close connections to Nazi-leaders
- Much more in M. Eckerts biography (to appear in 2016)

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Many thanks for your patience

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ENERGÍAS RENOVABLES

FUNDACIÓN CENER-CIEMAT

FINITE NUMBER OF BLADES EFFECT ON THE TIP AND THE ROOT

Task 11 IEA TEM on Aerodynamics
13th Jan 2016, NREL Boulder, USA

Xabier Munduate



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índice

1. Description of the rotor model
 2. Tangential (Azimuthal) induced velocity
 3. Circulation
 4. Conclusions
-



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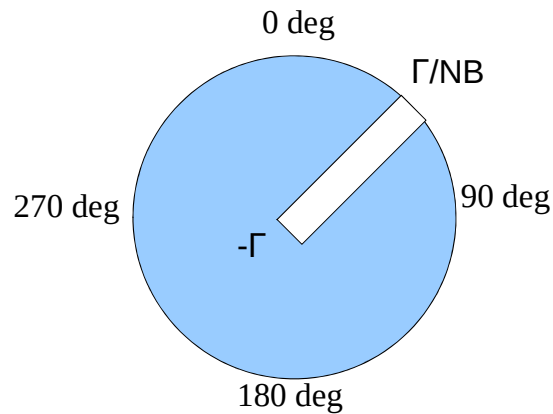


Description of rotor model

Joukowski Rotor with Constant Circulation. Positive circulation at the tip and negative at the root.



Simple vortex model, that follows the Joukowski concept of rotor with rectilinear filaments of circulation at the tip and the root of each blade.



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Description of rotor model

The effect of the finite number of blades has been study. The circulation (the tangential induced velocity V_θ) just behind the rotor, promoted by the root and tip rectilinear vortex filaments, has been calculated all along the span of the blade, from root to tip, following:

$$V_\theta = \Gamma / 4\pi r$$



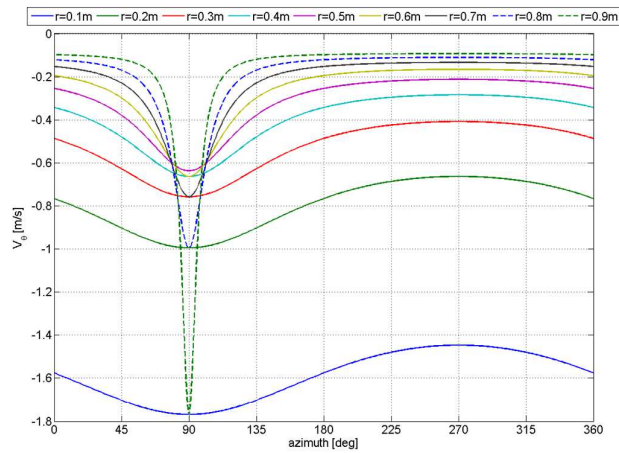
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1 BLADE ROTOR



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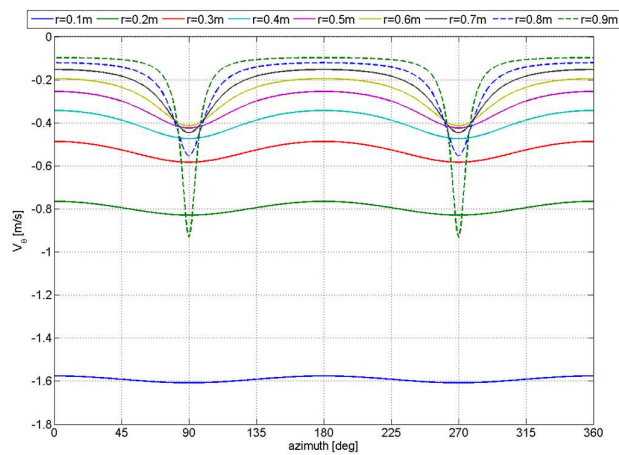
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2 BLADE ROTOR



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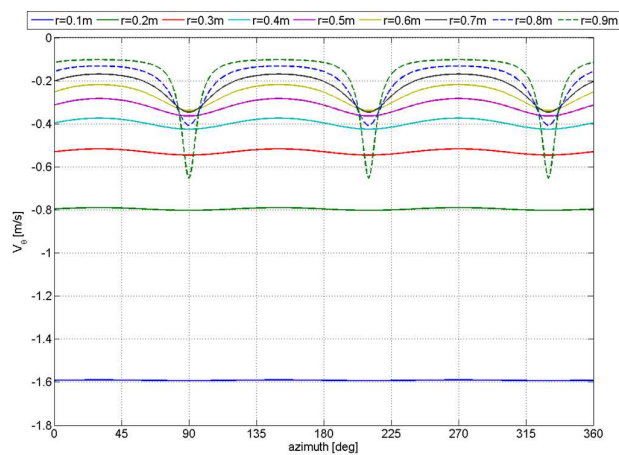
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CEAVAL



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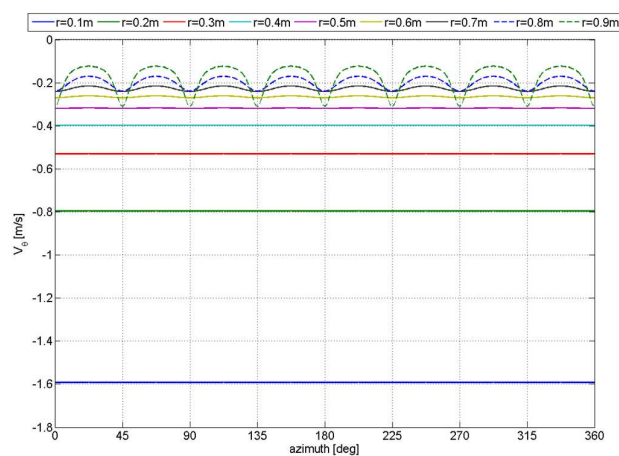
3 BLADE ROTOR



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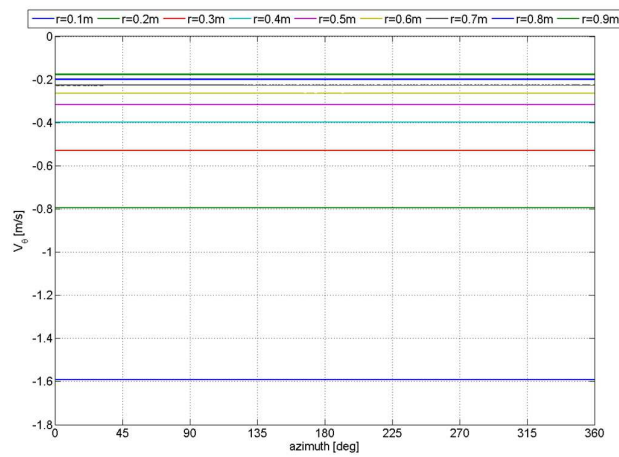
8 BLADE ROTOR



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100 BLADE ROTOR



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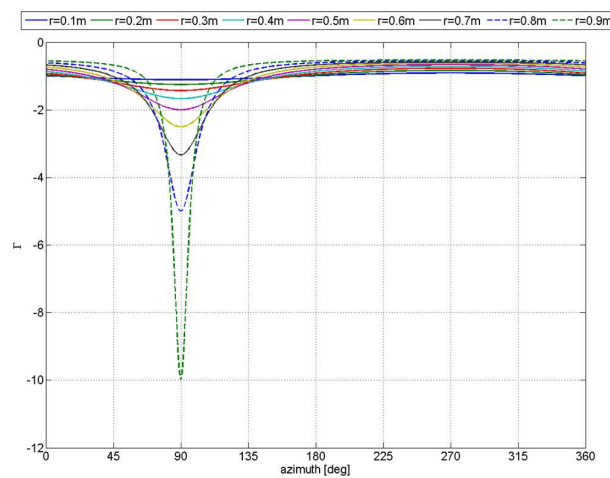
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1 BLADE ROTOR



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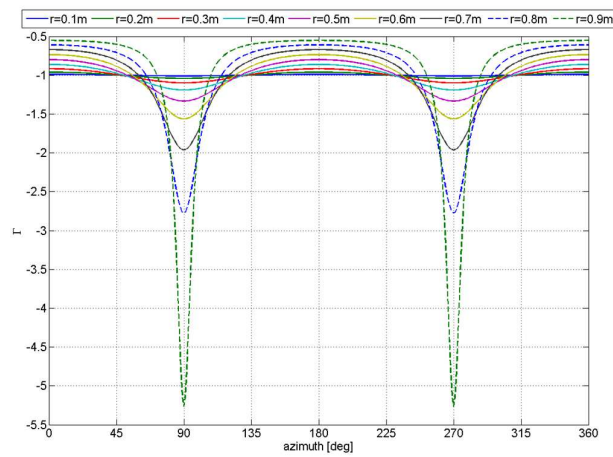
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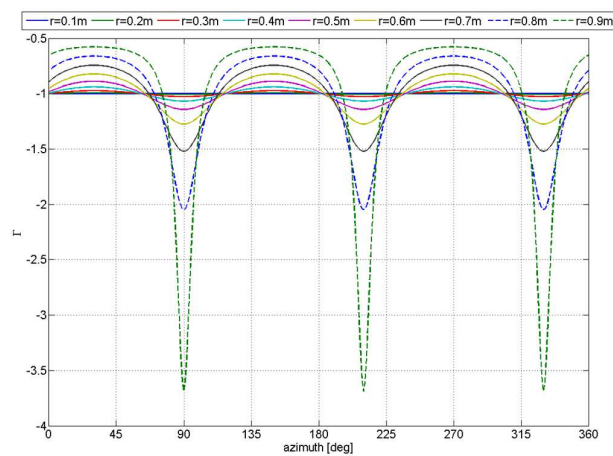
2 BLADE ROTOR



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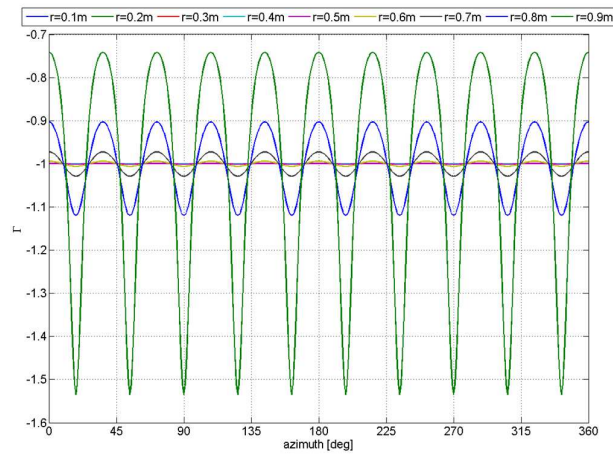
3 BLADE ROTOR



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10 BLADE ROTOR



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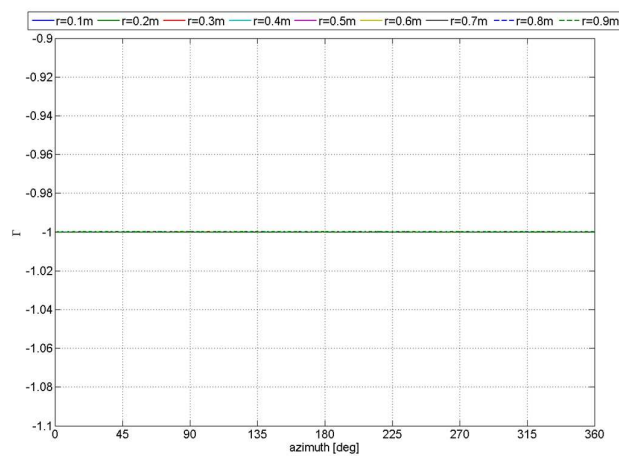
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100 BLADE ROTOR



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CONCLUSIONS

- The tangential induction variation on azimuth for 1B, 2B, 3B and ∞ B has been obtained.
- The main result is the appearance of the non axisymmetry of the flow, azimuthal variation of the tangential induction over a rotation from root to the tip for the finite number of blades rotor.
- An important result is the axisymmetry of the flow, no variation in azimuth of the induction, for the infinite number of blades case.



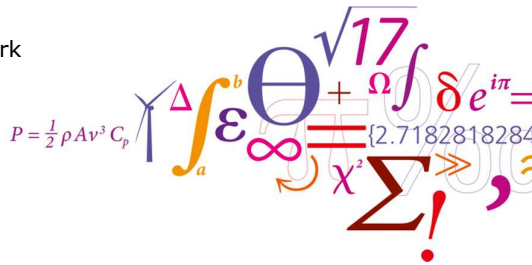
Wake flow characteristics at high wind speed



H.A. Madsen; T.J. Larsen; G.C. Larsen; K.S. Hansen

DTU Wind Energy
Technical University of Denmark
P.O. 49, DK-4000 Roskilde, Denmark

hama@risoe.dtu.dk



DTU Wind Energy
Department of Wind Energy

Acknowledgement



Simens Wind power is acknowledged for providing access to the experimental load data from the Lillgrund Wind Farm and for providing aeroelastic data input for their turbine.

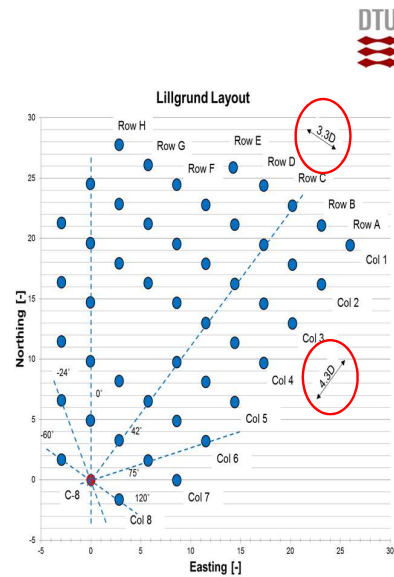
The presented work is part of a bigger study reported in the following references:

- Larsen, T. J., Larsen, G. C., Madsen, H. A., Thomsen, K. and Petersen, S. M. "Comparison of measured and simulated loads for the Siemens SWT 2.3 operating in wake conditions at the Lillgrund wind farm using HAWC2 and the Dynamic Wake Meander model," *EWEA offshore 2015*, Copenhagen, 10 – 12 March, 2015.
- Larsen, T. J., Larsen, G. C., Madsen, H. A., Thomsen, K. and Petersen, S. M. "Wake effects above rated wind speed. An overlooked contributor to high loads in wind farms," *EWEA 2015 Conference*, 17-20 November, Paris.
- Machefaux, E., Larsen, G. C., & Mann, J.. Multiple Turbine Wakes," *DTU Wind Energy PhD*; No. 0043(EN), 2015.
- Machefaux, E., Larsen, G. C. and Murcia Leon, J. P. "Engineering models for merging wakes in wind farm optimization applications". *Wake conference 2015*, Gotland.

Lillgrund Wind farm



- 48 SWT 2.3 MW PRVS turbine
- Lowest spacing is 3.3D
- Turbine C-8 is fully instrumented with strain gauges. We have access to results for the blade bending moment and tower bottom bending moment
- The MET mast data cannot be used for all sectors. Outerly placed turbines was used for wind speed measurement (power and pitch angle)



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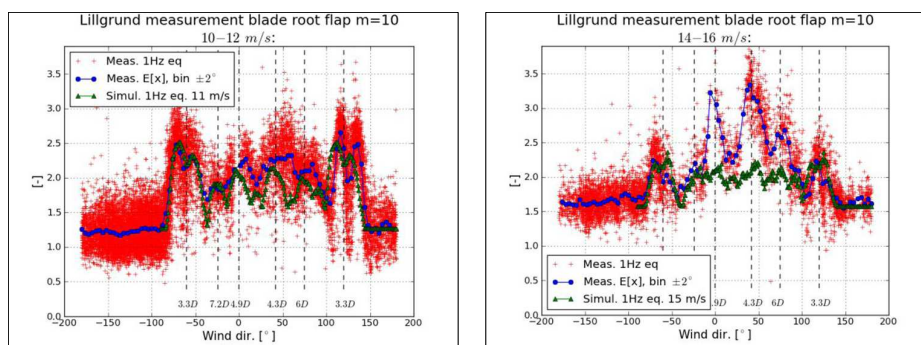
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January 13, 2016, NREL
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Motivation for study

- results from comparisons of measurements at the Lillgrund wind farm with simulations with the DWM model

T.J.Larsen et al. (2015a)

Flapwise fatigue loads normalized with loads at 9m/s

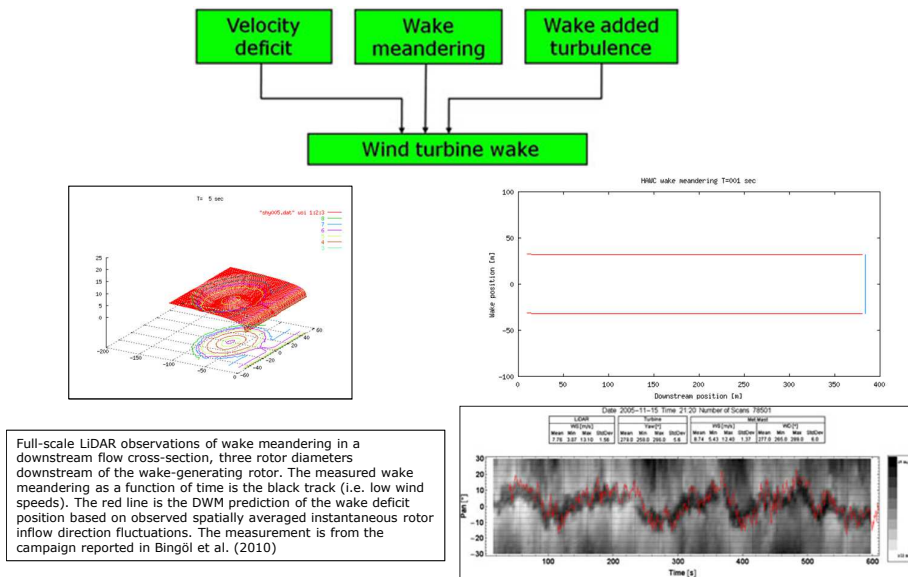


Load generation mechanism in DWM: meandering of a velocity deficit

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The DWM model



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Merging of wakes in the DWM model until 2015



T.J.Larsen et al. (2013)

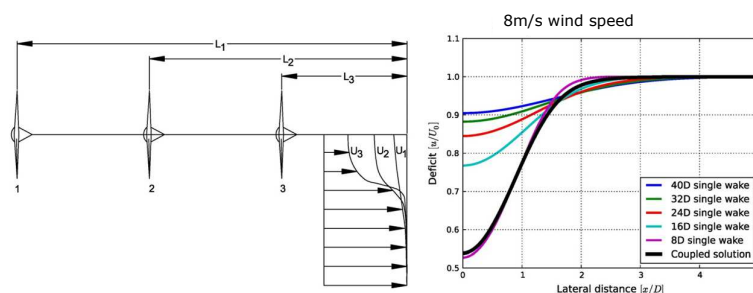


Figure 3. Illustration of the principles behind multiple wake deficits without effects of meandering. (Left) Turbine configuration showing the first three turbines in a row of five turbines and their individual deficits. (Right) Final individual deficits of the five upstream turbines separated with eight diameters as calculated with the axis-symmetric eddy viscosity model compared with a coupled solution including all turbines.

$$u_{\text{def,final}}(r) = \min(u_{\text{def,1}}(r), u_{\text{def,2}}(r), \dots, u_{\text{def,N}}(r)) \quad (2)$$

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Coupled solution approach

no meandering
turbine row aligned with wind direction

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Computational procedure



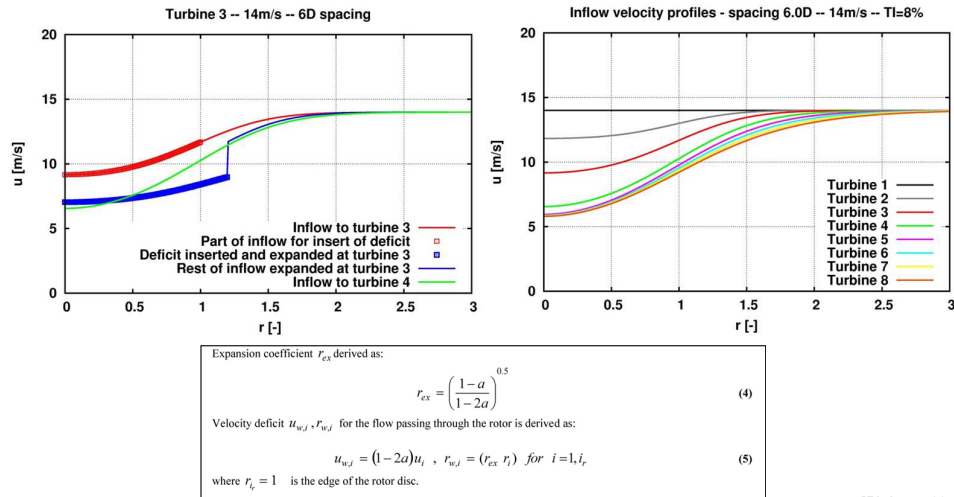
A sequential computation of the deficits in a row is only possible for the wind direction aligned with the row due to the axis-symmetric BLE model

1. the inflow profile is from the BLE model computation of the deficit from the turbine in front
2. mean wind speed over the rotor disc is derived
3. CT is determined from a look up table for the actual turbine based on wind speed
4. induction a is computed
5. deficit inserted and expanded based on the BEM wake solution
6. the deficit is input to the BLE model (axis-symmetric eddy viscosity model)
7. computation of the deficit development by the BLE model which gives the inflow to the next turbine
8. repeat all steps for next turbine

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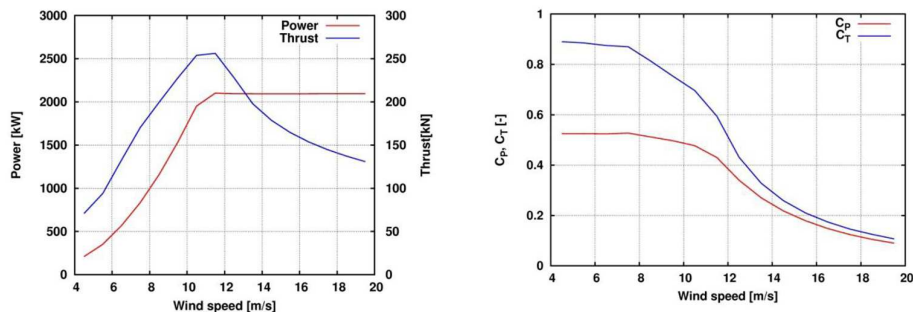
Example - computation of velocity deficit from turbine 3



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Power and thrust characteristics for the NM80 turbine are used



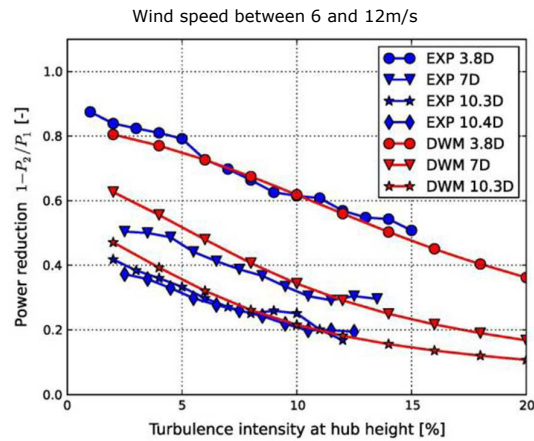
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The BLE model



Calibration of the eddy viscosity in the BLE model has previously been performed by comparison with power measurements in three wind farms - T.J.Larsen et al. (2013)



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Results

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Power characteristics

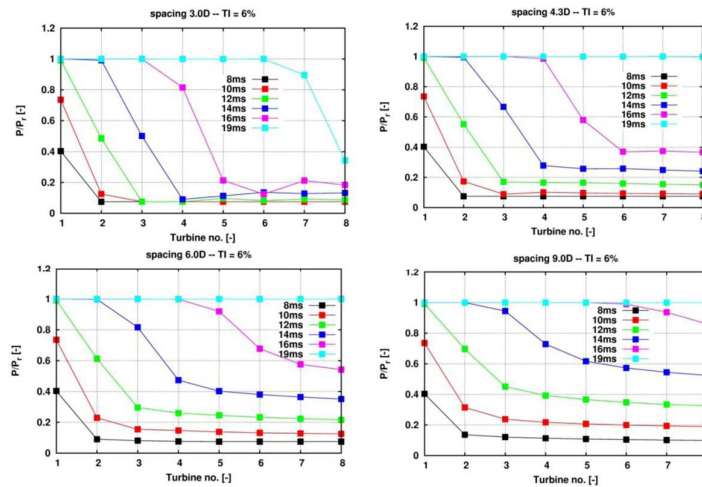


Figure 5 Variation of the turbine power non-dimensionalized with rated power along the row for different wind speeds and different turbine spacing's.

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Wind speed characteristics

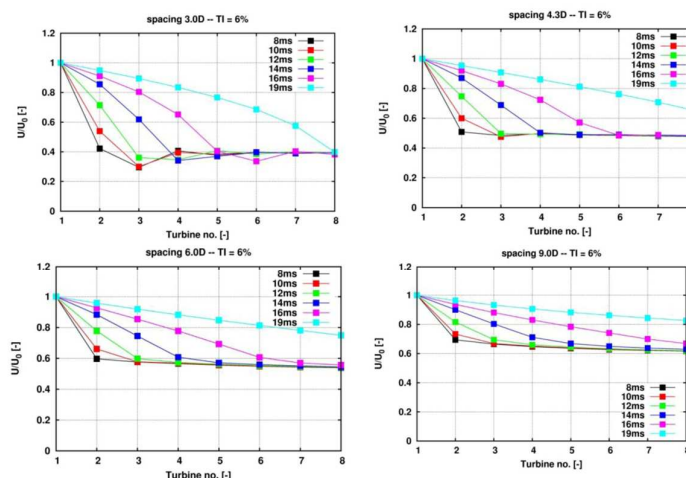
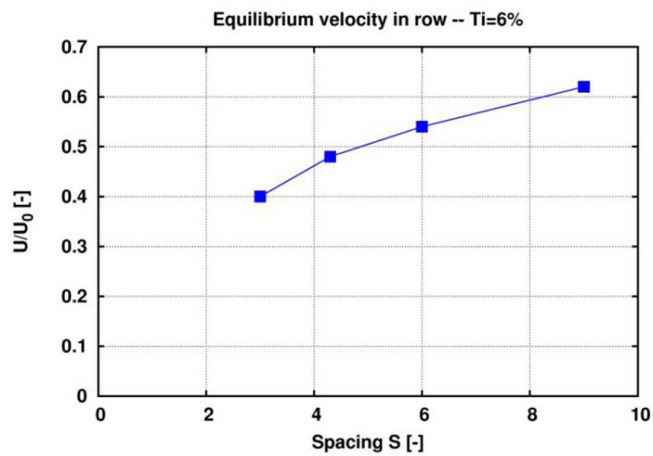


Figure 6 Variation of the mean inflow velocity (non-dimensionalized with free wind speed) over the rotor disc along the row for different wind speeds and different turbine spacing's.

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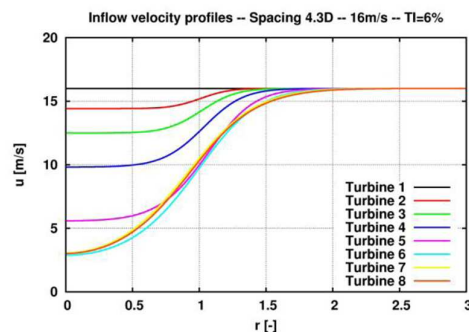
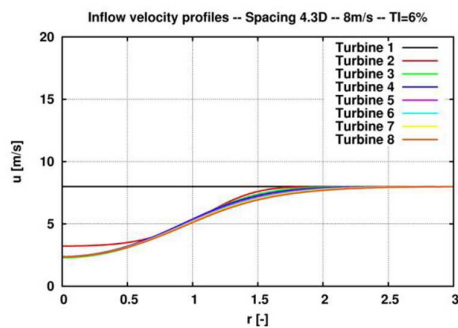
The equilibrium wind speed in a row



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Deficit characteristics at low and high wind speed



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Wake merging methods



SUM method

$$u(r) = U_0 \sum_{i=1}^N \left(1 - \frac{u_i(r)}{U_0} \right)$$

N is number of upstream wake affecting turbines

MIN method

$$u(r) = \min[u_i(r)]_{i=1}^N$$

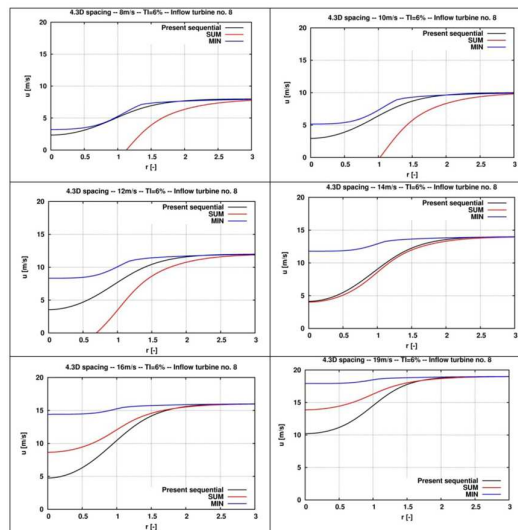


Figure 9 Comparison of two models SUM and MIN for merging of wakes at the last turbine in a row with 8 turbines and a spacing of 4.3D with the present sequential model.

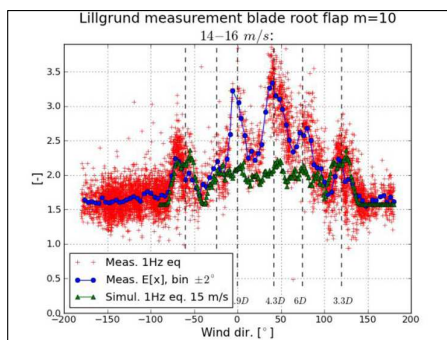
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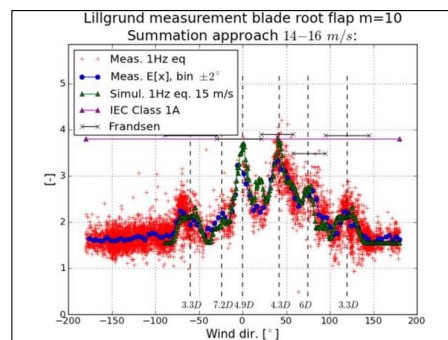
SUM merging method implemented in the DWM model



MIN method



*SUM method



* Larsen, T. J., Larsen, G. C., Madsen, H. A., Thomsen, K. and Petersen, S. M. "Wake effects above rated wind speed. An overlooked contributor to high loads in wind farms," *EWEA 2015 Conference*, 17-20 November, Paris.

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Conclusions



- ❑ A sequential, engineering model approach for computation of the flow aligned with a row of turbines has been used to study the wake flow characteristics below as well as above rated power.
- ❑ The simulations show that the last turbines in a row can experience inflow with a very strong velocity deficit that causes high loading.
- ❑ A so-called non-dimensional equilibrium wind speed is found at some distance down in the row, depending mainly on spacing and probably ambient turbulence intensity.
- ❑ The simulation results show that a summation of the deficits at high wind speed is a good choice, although still deviating for decreasing, increasing wind speed.

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**Thank you
for your attention!**

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AVATAR WP4

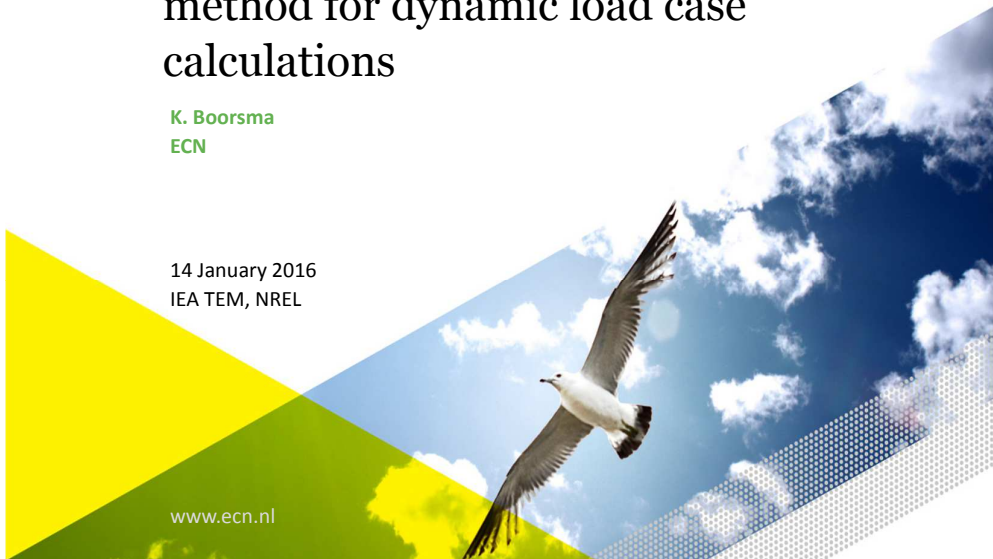
Application of a vortex line method for dynamic load case calculations



K. Boorsma
ECN

14 January 2016
IEA TEM, NREL

www.ecn.nl



Outline

- Introduction
- Set-up
 - Code description
 - Calculation settings
- Results
 - Yawed flow
 - Extreme transient shear
 - Half wake
- Conclusions



Source: Fotovlucht Soesterberg, Eric Vorstenbosch

Introduction



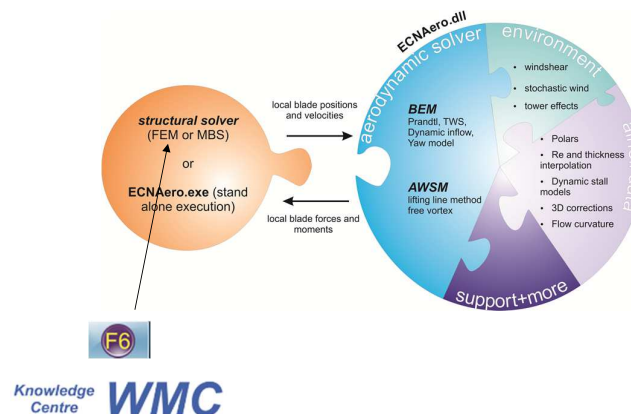
- AVATAR WP4
 - Aero-elastic calculations with emphasis on comparison between different aerodynamic models
 - INNWind and AVATAR reference turbines
- Several work packages
 - Stability analysis (frequency domain)
 - Standstill
 - Yawed flow
 - Extreme transient shear and half wake
 - Fatigue (turbulence)
 - And more

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Set-up: Code description



- ECN Aero Module coupled to FOCUS-PHATAS

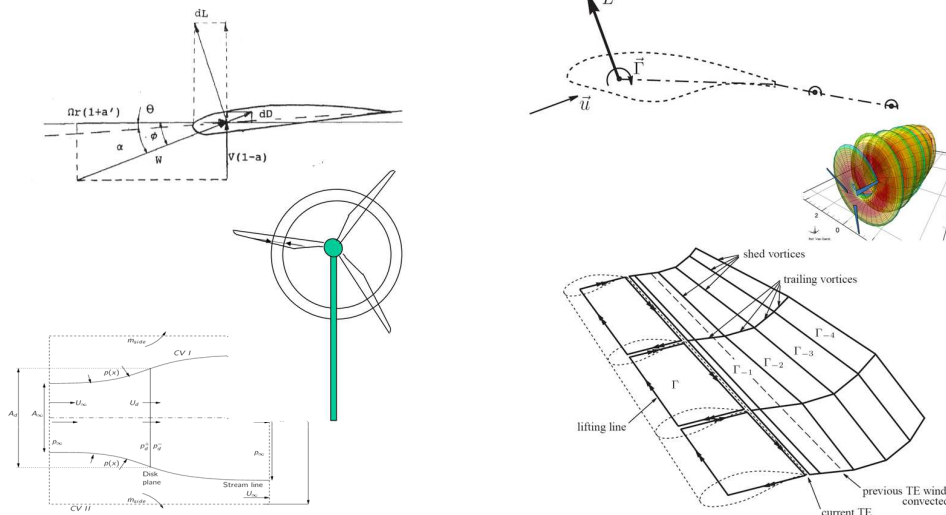


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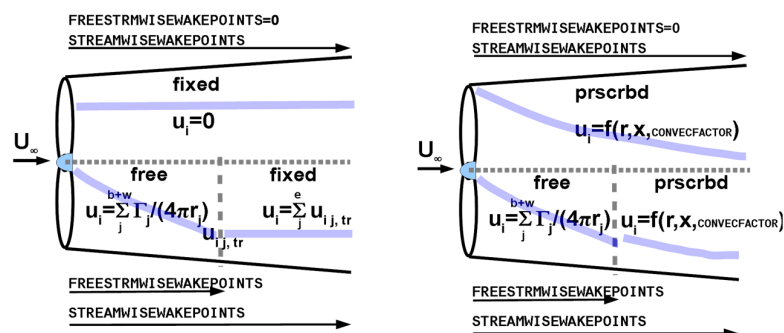
Set-up: Code description

- ECN Aero Module: BEM and AWSM



Set-up: Code description

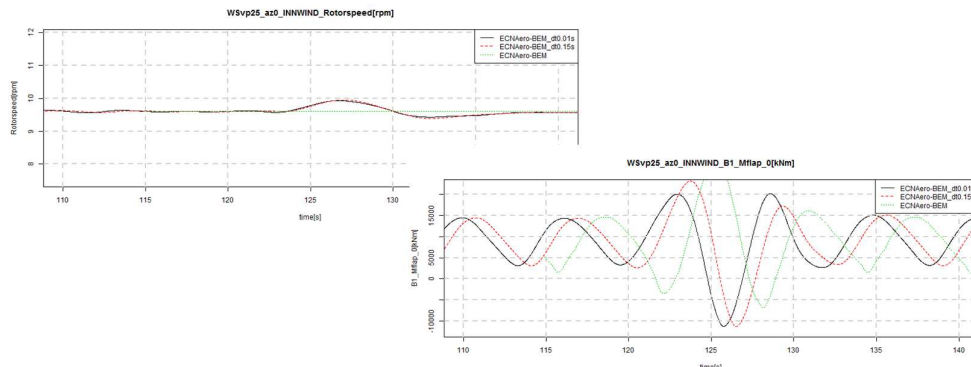
- AWSM wake convection
 - Free, fixed, prescribed or hybrid





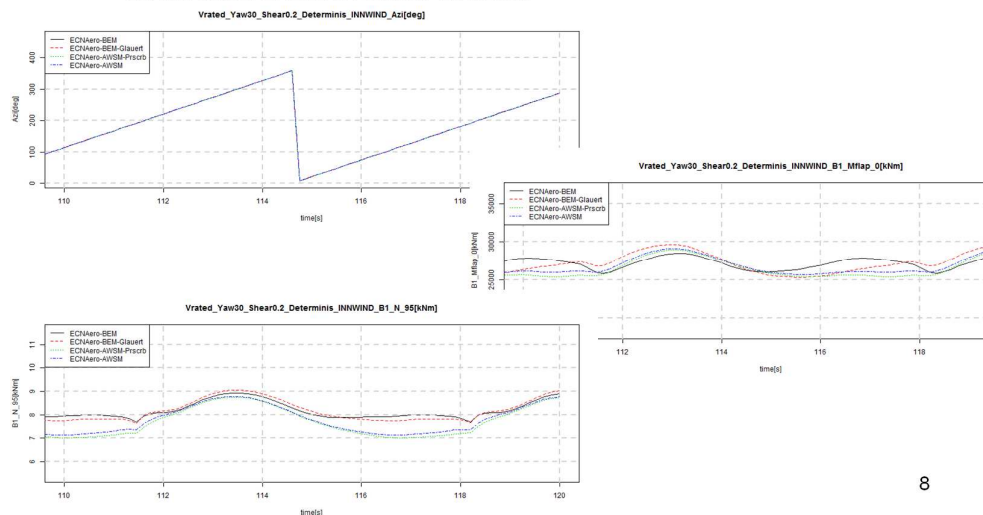
Set-up: Calculation settings

- Time step
 - Reduced to save CPU-time
 - Accuracy sacrifice limited according to BEM, depending on case
- Controller
 - For true comparison between aerodynamic models -> fixed pitch&rpm



Results: Yawed flow (30°)

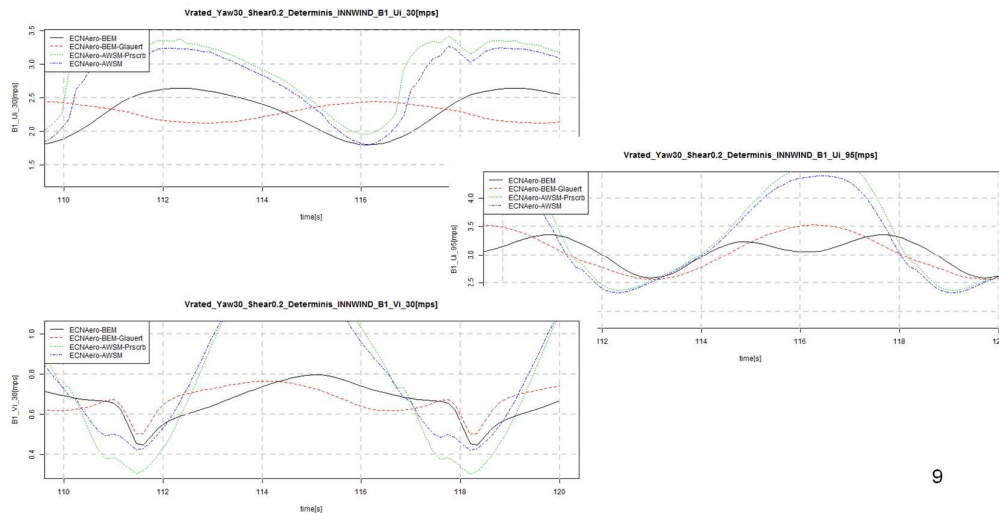
- Small differences in terms of loads



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Results: Yawed flow (30°)

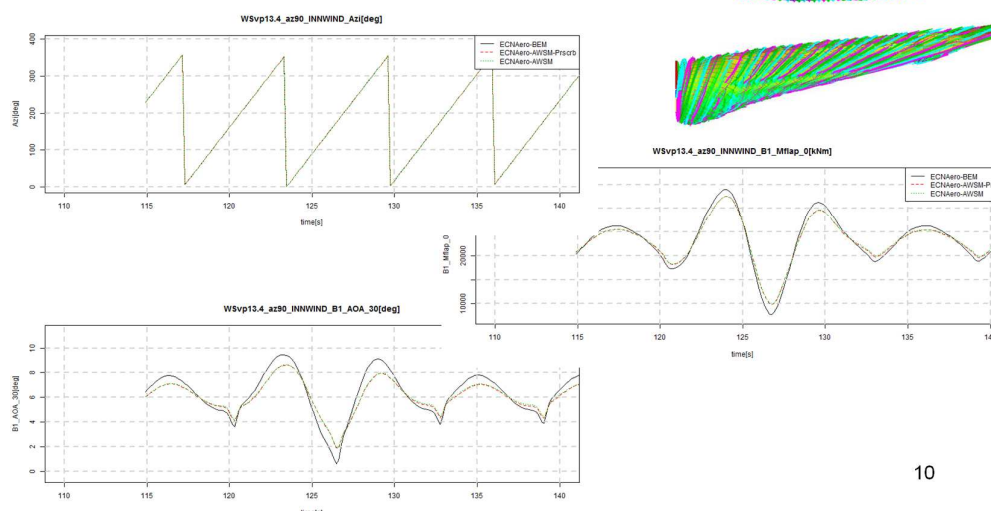
- Larger differences in terms of induced velocities



9

Results: Extreme transient shear

- INNWIND, rated power

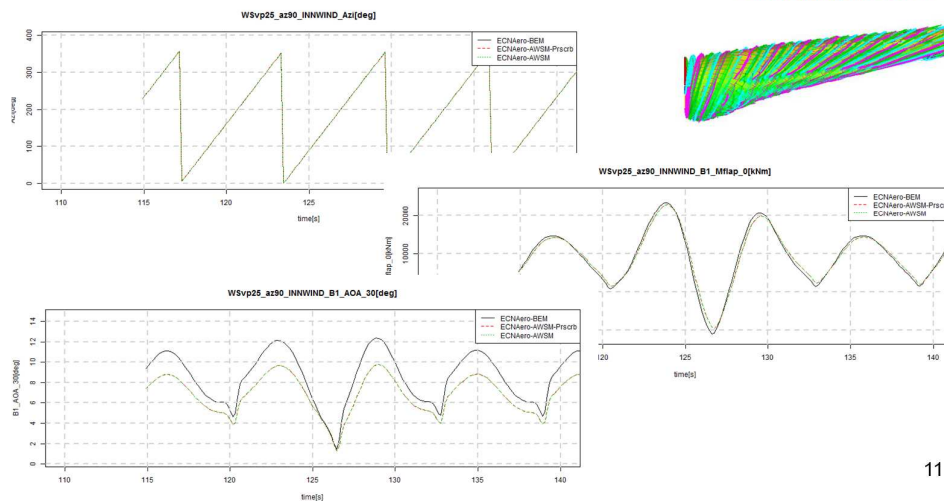


10



Results: Extreme transient shear

- INNWIND, cut-out

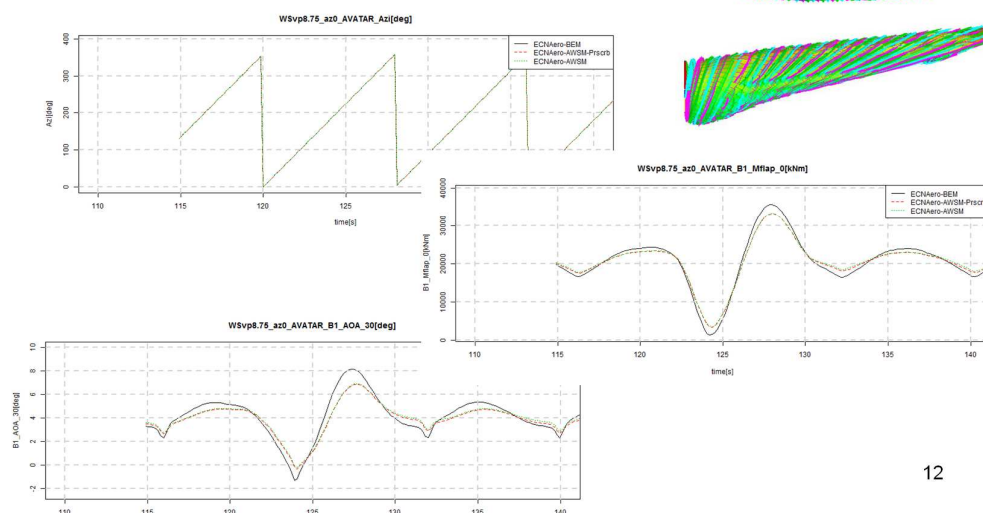


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Results: Extreme transient shear

- AVATAR, partial load

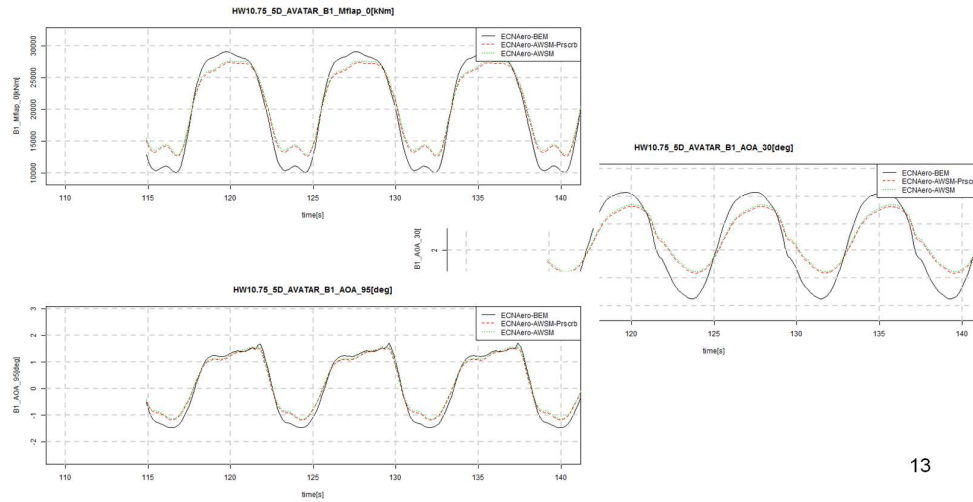


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Results: Half wake

- AVATAR, rated conditions

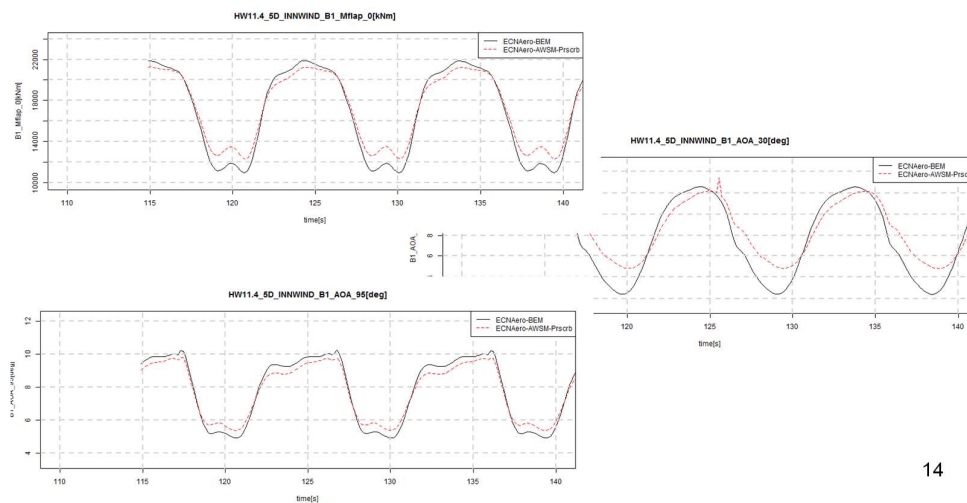


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Results: Half wake

- INNWIND, rated conditions

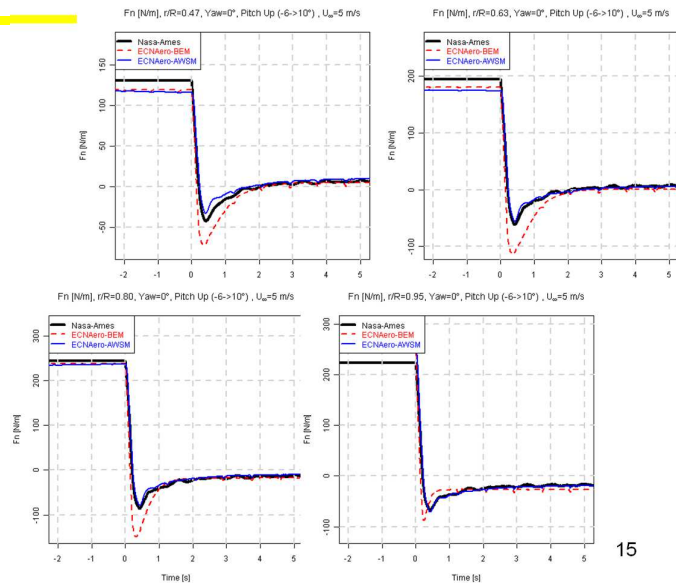


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Validation *NREL Phase VI – Dynamic inflow*



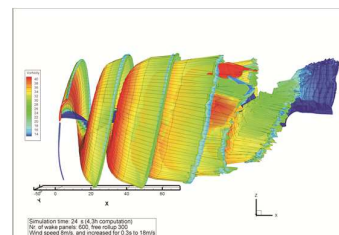
- Excellent agreement for AWSM
- Overshoot for BEM



15

Conclusions / future work

- Application of vortex line method is promising
 - Modeling the wake often damps fluctuation in case of large dynamics
 - Lower fatigue loads for selected cases?
 - Most relevant for cases with high induction -> INNWIND rotor
 - Note difference between BEM codes is often of same order..
- Evaluate other cases in coming months
 - Standstill (trailing vorticity)
 - Fatigue for stochastic winds
- Improve AWSM CPU performance
 - Parallelization (MPO)
 - Separate update frequencies between local airfoil aero and vortex shedding

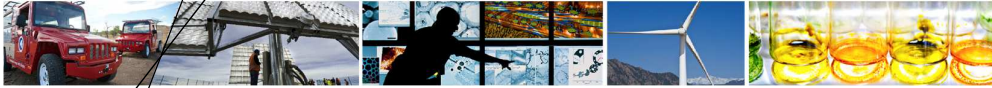


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Using Siemens 2.3MW experimental data for FAST verification and validation



Srinivas Guntur, Jason Jonkman, Scott Schreck, Bonnie Jonkman, Qi Wang and Mike Sprague

National Renewable Energy Laboratory, Golden, CO

Michael Hind and Ryan Sievers

Siemens Energy Inc., Boulder, CO

13 Jan., 2016.

IEA Task 11 - Topical Expert Meeting on Aerodynamics

NWTC/NREL, CO, USA

NREL is a national laboratory of the U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy, operated by the Alliance for Sustainable Energy, LLC.

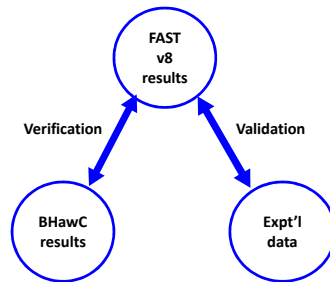
Introduction

- **What?**
 - Use of the Siemens 2.3MW turbine experimental data to perform verification and validation of FAST v8.
- **Why?**
 - To highlight the improvements of the latest version of FAST over its ancestor versions.
- **How?**
 - Compare results from simulations with experimental data as well as the results from Siemens in-house aeroelastic simulation tool BHawC.
 - Cases selected based on IEC 61400-13.

FAST v8 Verification and Validation for a MW-scale Wind Turbine with Aeroelastically Tailored Blades. Guntur et al., AIAA Scitech, Jan 2016, San Diego, CA.

Objective

- To perform a three-way **code-to-code** and **code-to-measurements** comparison
- To investigate the improvements provided by the latest version of FAST.



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Experiments

NWTC @ NREL



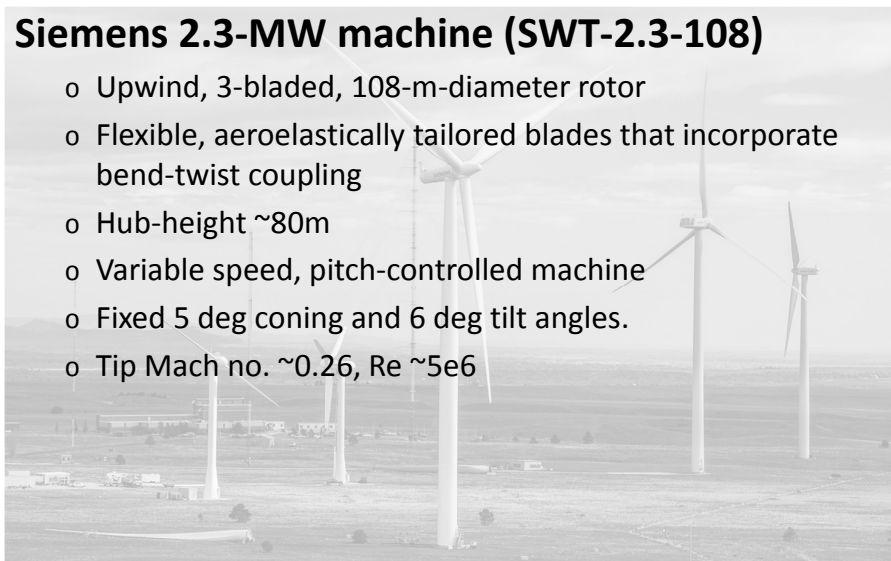
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Experimental measurements

Siemens 2.3-MW machine (SWT-2.3-108)

- o Upwind, 3-bladed, 108-m-diameter rotor
- o Flexible, aeroelastically tailored blades that incorporate bend-twist coupling
- o Hub-height ~80m
- o Variable speed, pitch-controlled machine
- o Fixed 5 deg coning and 6 deg tilt angles.
- o Tip Mach no. ~ 0.26 , $Re \sim 5e6$



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Experimental measurements

SWT-2.3-108 instrumentation

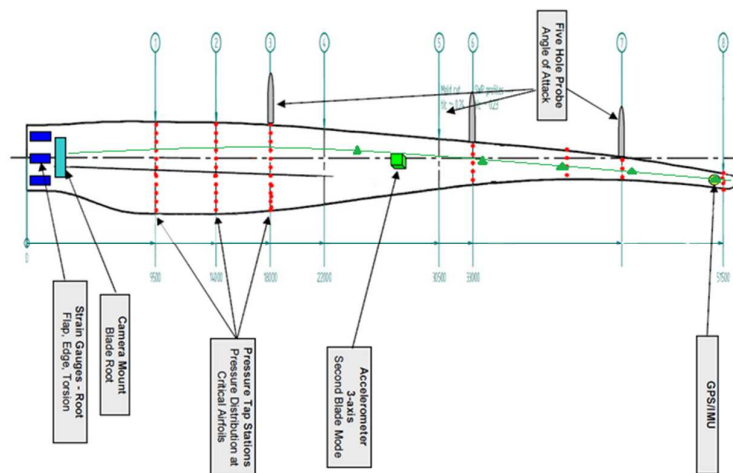
- o Strain-gages at the blade root, tower top, tower bottom, and main shaft
- o FiberBragg strain sensors along the blade
- o Data acquisition system to record rotor speed and electrical power
- o Blade surface pressure taps (7 locations), pitot tubes (3 locations)
- o Inflow data recorded using a 135-m meteorological tower located approx. 2.5 diameters upstream
- o Data recorded at 100Hz, and packaged into 10-min time series

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Experimental measurements

Instrumented blade – schematic



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Experimental measurements

- Large amount of data collected over a long period of time – 2013 to 2015
- A total of 1141 10-min data sets were identified covering a range of inflow wind speeds and turbulence intensities, using the IEC 61400-13 test matrix as a guideline.

Turbulence Intensity [%]	Mean Wind Speed [m/s]														
	Region 2								Region 3						
	3.5-4.5	4.5-5.5	5.5-6.5	6.5-7.5	7.5-8.5	8.5-9.5	9.5-10.5	10.5-11.5	v (rated) 11.5-12.5	12.5-14	14-16	16-18	18-20	20-22	22-24
<3	0	4	6	3	0	0	0	0	0	0	0	0	0	0	0
3-5	0	20	19	19	9	5	1	1	0	0	0	0	0	0	0
5-7	0	32	50	32	14	3	7	3	2	2	0	0	0	0	0
7-9	0	43	39	44	17	17	3	7	2	2	2	0	1	0	1
9-11	0	32	42	35	29	13	7	11	9	5	9	5	0	0	1
11-13	0	22	21	16	16	9	12	15	16	14	15	6	4	1	0
13-15	0	2	16	25	10	6	14	15	24	11	15	14	5	1	0
15-17	0	1	10	11	6	4	7	12	12	10	18	8	9	2	0
17-19	0	2	12	6	0	2	3	12	4	10	8	7	5	0	0
19-21	0	2	3	3	2	1	3	1	4	2	3	1	0	0	0
21-23	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0
Sum	0	160	218	195	103	60	57	77	73	57	70	41	24	4	2



Modelling SWT-2.3-108

FAST v8 model of SWT-2.3-108

- The tower, blades and the main shaft are modeled as elastic bodies with properties obtained from Siemens, and the tower base was fixed.
- A controller obtained from Siemens in a Bladed-style DLL form was used for pitch and torque control.
- Aerodynamic properties were obtained from Siemens. The lift-data was corrected for rotational augmentation using the Snel model, consistent with the BHawC simulations.
- Mann turbulent boxes used in both BHawC and FAST simulations. Shear was modelled using the IEC power law coefficient of 0.2.
- The tower, nacelle, and drive train were modelled in ElastoDyn, while the blades were modelled in BeamDyn as well as ElastoDyn.

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BHawC model of SWT-2.3-108

- The same 1141 cases were also simulated using BHawC; results were shared via a cooperative research and development agreement between NREL and Siemens.
- All input parameters almost identical in FAST and BHawC simulations. Only differences between the models: internal solvers, wind shear and controller.
- BHawC simulations were 11-min long, and the first minute was ignored for any transients.

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Methods

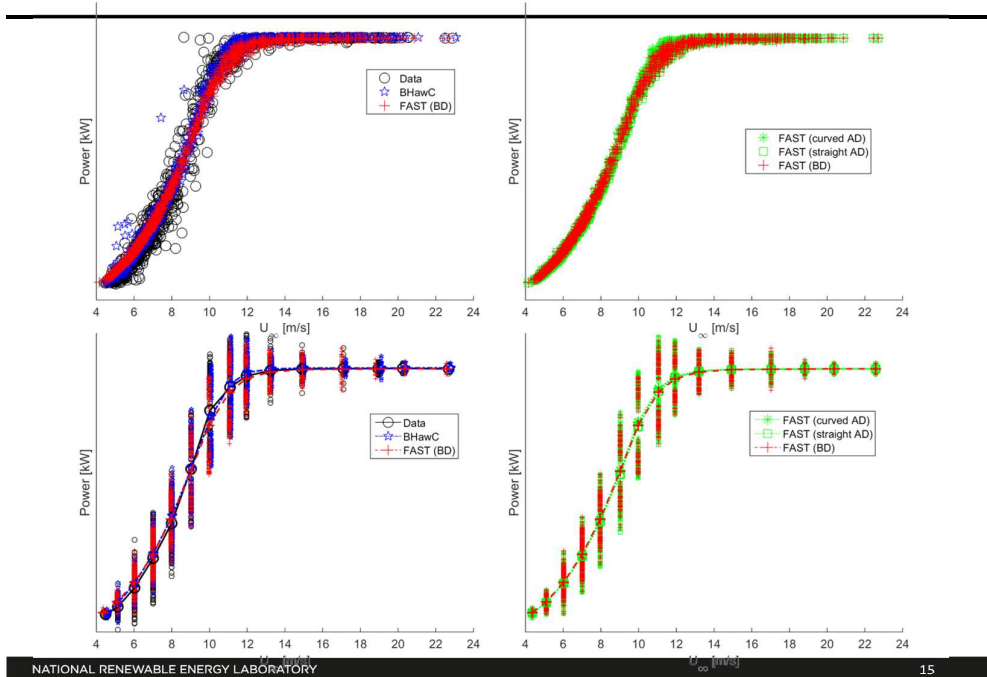
- **Examine improvements to FAST v8:**
 - Simpler structural model (ElastoDyn) compared to new non-linear high-fidelity beam model (**BeamDyn**) - see Wang et al. (AIAA 2016 162-WE-3)
 - Straight aerodynamic blades (AeroDyn v14) compared to aerodynamic blades curved in 3D space (**AeroDyn v15**) – see Damiani et al. (AIAA 2016 213-WE-4)
- **Three sets of FAST simulations were carried out for each 10-min case:**
 - FAST with straight ElastoDyn blades and straight AeroDyn blades
 - FAST with straight ElastoDyn blades and curved AeroDyn blades
 - FAST with curved BeamDyn blades and curved AeroDyn blades

This way, relative improvements between the previous and new capabilities of FAST are investigated.



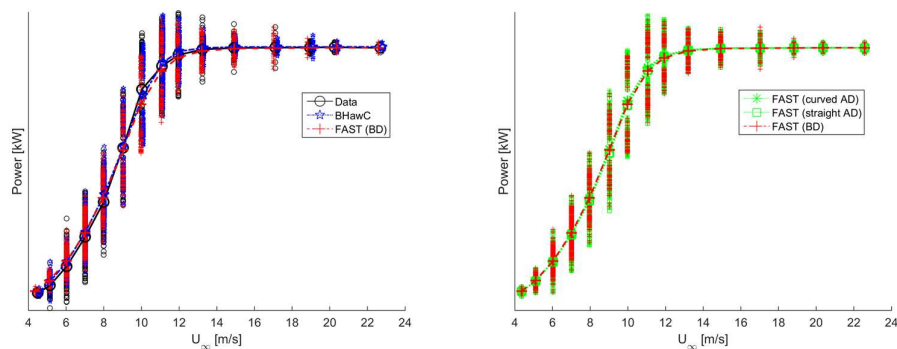
Results

Verification & Validation - Results



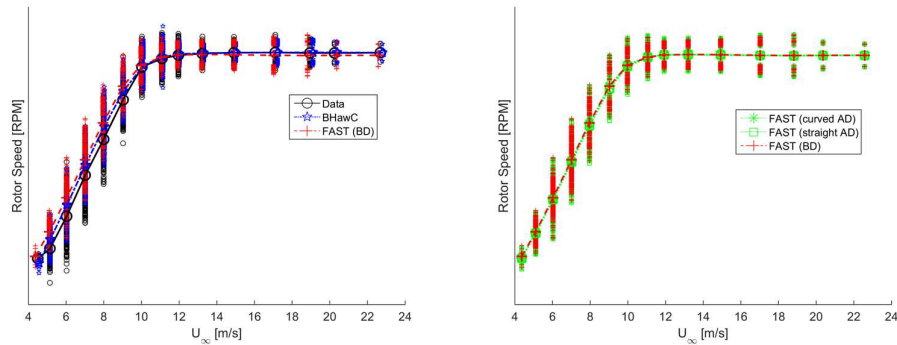
Verification & Validation - Results

Power



Verification & Validation - Results

Rotor Speed

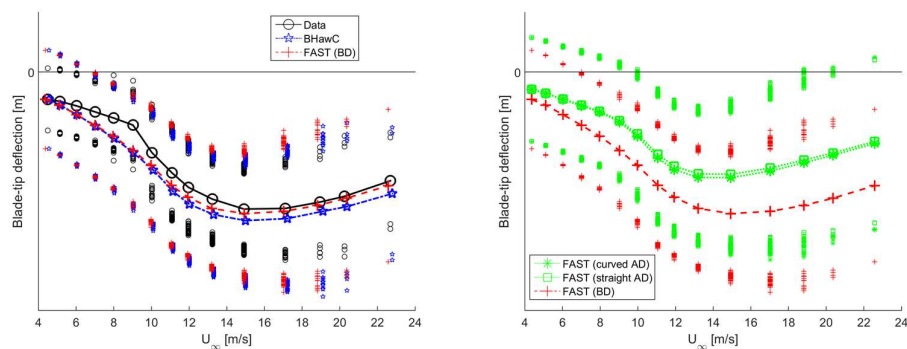


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Verification & Validation - Results

In-plane blade-tip deflection

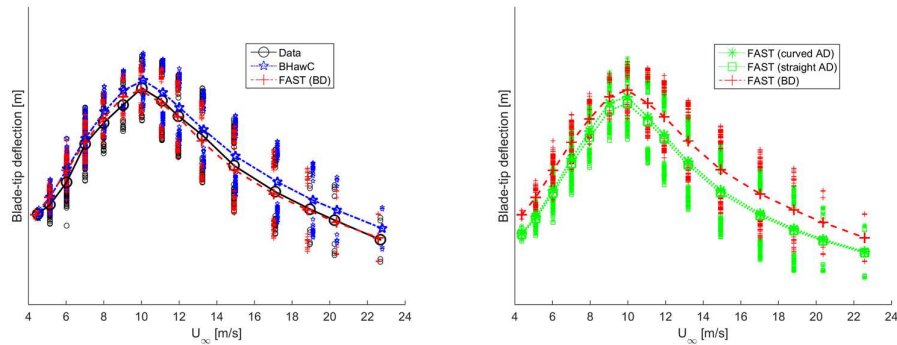


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Verification & Validation - Results

Out-of-plane blade-tip deflection

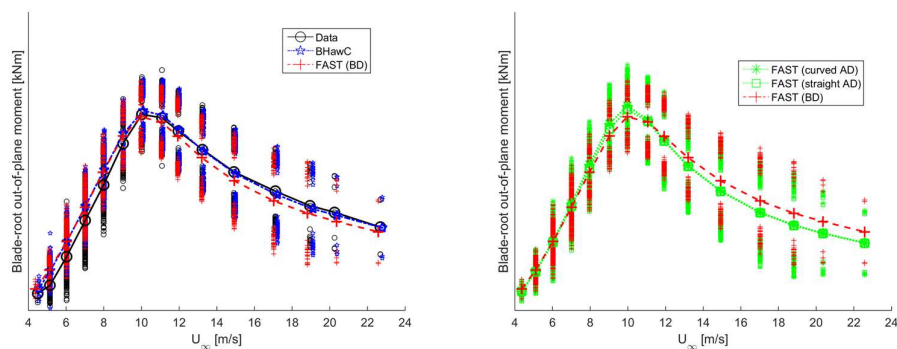


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Verification & Validation - Results

Blade-root out-of-plane bending moments

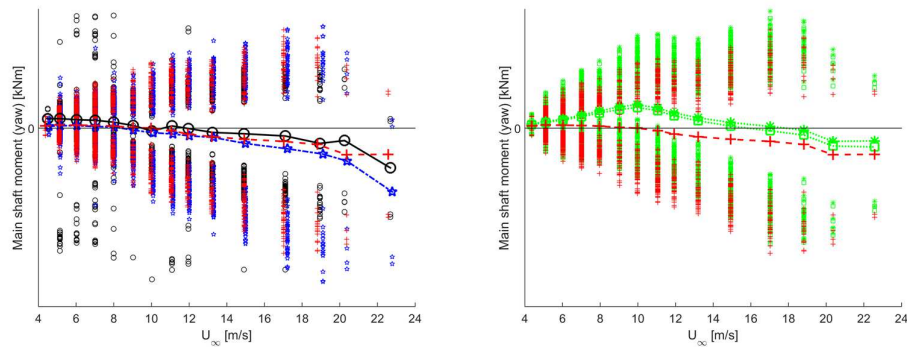


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Verification & Validation - Results

Main shaft bending moments (yaw, non-rotating)

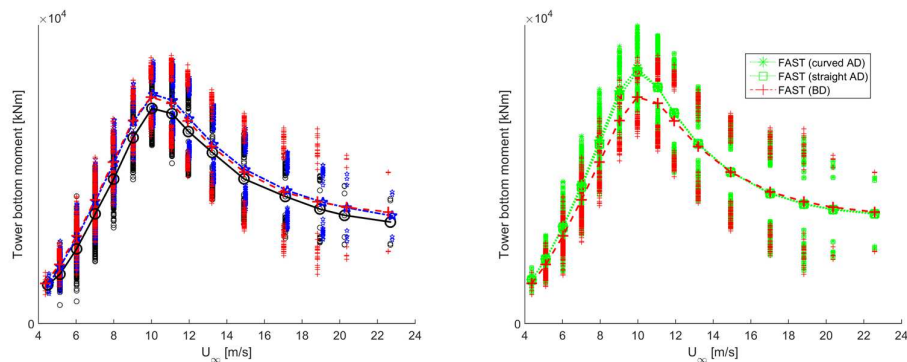


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Verification & Validation - Results

Tower-bottom fore-aft moment



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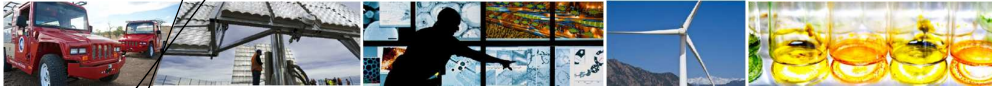
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Conclusions

- A three-way comparison is performed between results from FAST simulations, results from BHawC simulations, and experimental measurements.
- Overall, FAST with BeamDyn shows significant improvement over ElastoDyn
- Improvement is primarily in the blade-tip deflection, but this improvement also propagates into other quantities of interest
- Use of straight vs curved aerodynamic blade shows no significant impact
- The experimental data is generally of very high quality, however some uncertainties remain in determining the accuracy of some of the channels.
- Minor discrepancies between FAST and Siemens' data may be related to the controller or the shear model.

Future work

- A more detailed analysis with PSDs of the loads.
- More specific aerodynamic analysis using pressure data.
- Uncertainty analysis of the experimental data acquisition process.
- Sensitivity analysis of modeling a full wind turbine.
- Simulation of wider range of conditions (extreme events, start up, shut down, high yaw errors, etc.)



Thank you for your attention!

FAST v8: What's new?

- **New modular framework, comprising of**
 - ElastoDyn: Structural model
 - **BeamDyn**: The new nonlinear blade model
 - ServoDyn: Control and electrical drive dynamics module
 - **AeroDyn**: The aerodynamics module
 - InflowWind: Wind inflow characterization module

FAST v8: What's new?

- **BeamDyn: The new nonlinear finite element blade model supporting**
 - large deflections, with bending, torsion, shear, and extensional degrees of freedom (DOF)
 - anisotropic composite material couplings
 - a reference axis that permits blades that are not straight (built-in curve, sweep, and sectional offsets)
- **AeroDyn v15: The new aerodynamics module that**
 - fixes underlying problems with the original theoretical treatments
 - introduces improved BEM algorithm with skewed-wake and unsteady airfoil aerodynamics models*
 - enables modeling of highly flexible and non-straight blades



www.eera-avatar.eu

AVATAR project Advanced Aerodynamic Tools for lARge Rotors

Gerard Schepers
January 13th, 2016

IEA Topical Expert meeting
NREL USA

This project has received funding from the European Union's Seventh Programme for research, technological development and demonstration under grand agreement No FP7-ENERGY-2013-1/n° 608396 .



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1) Acknowledgement G. Sieros, M. Stettner

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And many others

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2



EU FP7 Project initiated by EERA *)

1. Energy Research Centre of the Netherlands, ECN (Coordinator, G. Schepers, F. Savenije, O. Ceyhan)
2. Delft University of Technology, TUDelft C. Simao Ferreira
3. Technical University of Denmark, DTU N. Sørensen
4. Fraunhofer IWES B. Stoevesandt
5. University of Oldenburg, Forwind M. Wächter, H. Heisselmann, M. Holling
6. University of Stuttgart, USTUTT T. Lutz, E. Jost
7. National Renewable Energy Centre, CENER A. Gonzalez
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10. National Technical University of Greece, NTUA S. Voutsinas
11. Politecnico di Milano, Polimi A. Croce
12. GE Global Research, Zweigniederlassung der General Electric Deutschland Holding GmbH, GE M. Stettner, H.J. Kooijman
13. LM Wind Power, LM J. Madsen, C.F. Andersen

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*) European Energy Research Alliance ³



Period

- Project period:

November 1st 2013- November 1st 2017

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Main motivation for AVATAR: Aerodynamics of large wind turbines (10-20MW)

UPSCALING towards 10-20 MW turbines is expected to lead to turbines with:

- ☐ Low induction
- ☐ Long slender blades
- ☐ Thick airfoils
- ☐ High tip speeds
- ☐ Passive (e.g. vortex generators (vg's)) and active flow devices (e.g. flaps)

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Motivation, ctd

- We simply don't know if present aerodynamic models are good enough to design 10MW+ turbines
 - “No mature industry will ever design a MEuro machine with unvalidated tools”
- 10MW+ rotors violate assumptions in current aerodynamic tools, e.g.:
 - Reynolds number effects,
 - Compressibility effects
 - Flow transition and separation,
 - (More) flexible blades
 - Flow devices
- Hence 10MW+ designs fall outside the validated range of current state of the art tools.

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Avatar: Main objective

To bring the aerodynamic and fluid-structure models to a **next level** and **calibrate** them for all relevant aspects of large (**10MW+**) wind turbines

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Avatar: Work procedure

- **Problem:** No **10 MW** turbines are on the market yet for validation
- Hence: Validate **submodels** against experiments
 - **Pressurized** HDG tunnel of German Dutch Wind Tunnel facilities (DNW)
 - Airfoil measurements at Reynolds numbers up to **RE = 15 M** and low Mach (< 0.2)
 - LM: Wind tunnel airfoil measurements also at dynamic conditions
 - Forwind: Wind tunnel airfoil measurements at **representative turbulence**
 - TUDelft: Wind tunnel experiments on airfoils with **vortex generators, flaps**
 - NTUA: Wind tunnel experiments on airfoils with/without **vortex generators**
 - DTU : Danaero: Aerodynamic **field** experiments on a **2.3 MW** turbine and supporting 2D wind tunnel measurements
 - Note: Several experiments are supplied **in-kind**

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Avatar: Work procedure

Use the different models from partners in the project

- o It is a **cooperation** project!
- o In the project we have many models which range from computational efficient '**engineering**' tools to **high fidelity** but **computationally expensive** tools
- o Engineering tools are needed in **industrial** design codes
 - *Both LM and GE stated very clearly: We need engineering models in wind turbine aerodynamics!*
- o **High fidelity** models (**and intermediate** models) feed information towards **engineering** models

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Avatar: Work procedure

- **Demonstrate the value of the improved tools on 10 MW reference rotors with and without flow control devices**
 1. INNWIND.EU reference rotor (more or less **conventional** design philosophy)
 2. AVATAR reference rotor which should be more **challenging** from an aerodynamic point of view (e.g. lower induction, longer, more slender blades, thicker airfoils, higher tip speed).
- **Compare results from 'old' and improved models at the end of the project**

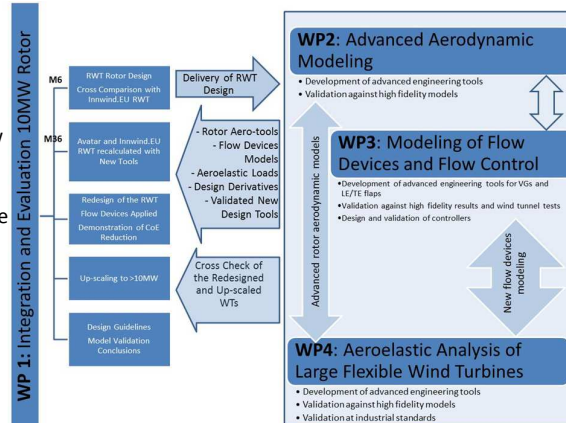
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Work Packages/ Work Package dependency

- WP1: Integration and evaluation of 10 MW RWT (CRES)
- WP2: Advanced aerodynamic modelling (DTU)
- WP3: Modelling of flow devices and flow control (CENER)
- WP4: Aeroelastic analysis of large flexible blades (NTUA)



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- And many others**

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Design of AVATAR Reference turbine

- AVATAR reference turbine based on INNWIND.EU reference turbine using its load envelope as constraint
- INNWIND.EU RWT characteristics:
 - Specific power of 401 W/m^2
 - Rotor diameter of 178.4 m
 - Rated rotor speed of 9.6 rpm and a tip speed of 90 m/s ,
 - Rated wind speed of 11.4 m/s
 - Hub height of 119 m → clearance to sea level of 30 m
 - The axial induction factor along the blade ~ 0.3 .
 - Airfoil family FFA-W3-xxx, $t/c = 0.24$ at tip

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Design of AVATAR RWT

Starting from Innwind.EU RWT values, but

- ❑ Specific power from $400 \text{ W/m}^2 \rightarrow 300 \text{ W/m}^2$
- ❑ D from $178.84 \text{ m} \rightarrow 205.8 \text{ m}$
- ❑ Rated wind speed from $11.4 \rightarrow 10.375 \text{ m/s}$
- ❑ Tip speed $90 \text{ m/s} \rightarrow 103.4 \text{ m/s}$ (keeping $\Omega = 9.8 \text{ rpm}$)
- ❑ Axial induction $\sim 0.3 \rightarrow \sim 0.24$
- ❑ Airfoil family based on DU airfoils, thickness of airfoils similar to Innwind.EU RWT

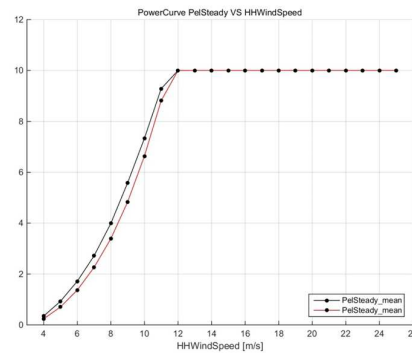
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Design of AVATAR RWT

- AVATAR turbine led to a 5% increase in energy production due to larger diameter where key load levels are maintained
- Note: Cost reduction for AVATAR turbine placed in a 1 GW wind farm is calculated in presentation at EERA-Deepwind also because of lower wake effects which are associated to lower induction
 - State of the art modelling of wind farm effects (wakes and electrical effects)
 - State of the art cost modelling



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Design of AVATAR RWT

- The operational conditions

Section Thickness	Re (rated)	Ma (rated)	Re (Min)	Ma (Min)
60.0%	7.0×10 ⁶	0.05	4.4×10 ⁶	0.03
40.1%	11.0×10 ⁶	0.07	7.0×10 ⁶	0.05
35.0%	14.0×10 ⁶	0.09	9.0×10 ⁶	0.06
30.0%	17.0×10 ⁶	0.12	10.0×10 ⁶	0.07
24.0%	20.0×10 ⁶	0.16	12.0×10 ⁶	0.10
24.0%	16.0×10 ⁶	0.25	11.0×10 ⁶	0.15
24.0%	13.0×10 ⁶	0.30	8.0×10 ⁶	0.18
21.0%	20.0×10 ⁶	0.16	12.0×10 ⁶	0.10
21.0%	16.0×10 ⁶	0.25	11.0×10 ⁶	0.15
21.0%	13.0×10 ⁶	0.30	8.0×10 ⁶	0.18

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Measurements in DNW-HDG pressurized tunnel

- Airfoil: DU00-W-212
- DNW-HDG:
 - Reynolds numbers up to 15 M
 - Pressure distribution measurements
 - Wake rake for drag determination, sidewise traversabel
 - Estimation of transition location using embedded Kulite pressure sensors as high frequency microphones
 - Fluorescent oil flow visualization
 - **Data were brought into a blind test and simulated by several parties also outside the project**

Note: DU00-W-212 is also measured by LM up to $Re=6M$ and by Forwind at controlled turbulent conditions up to 1M

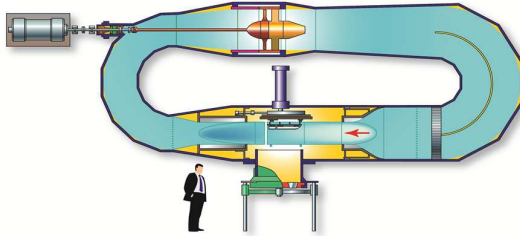
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DNW-HDG model, $c=15$ cm





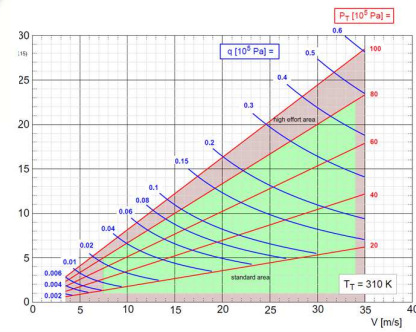
DNW-HDG Wind Tunnel



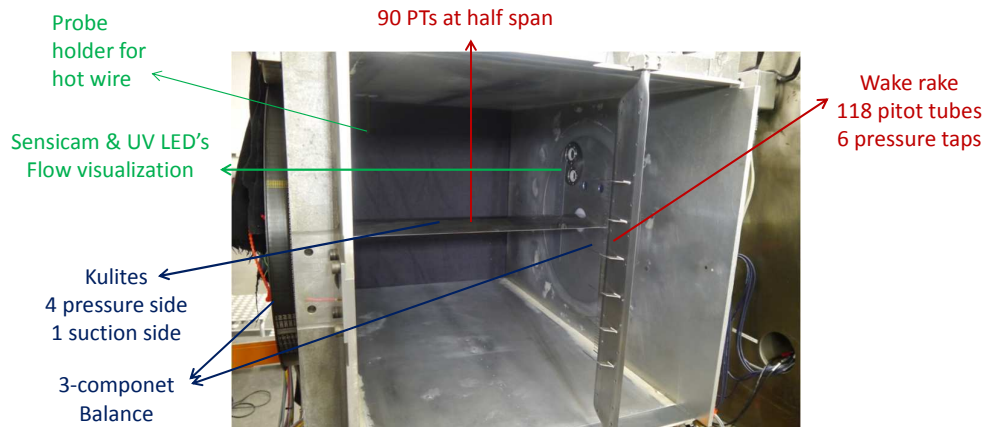
General Tunnel Characteristics

Test section : 60cmx60cm
 Fan Rpm : 200 – 820 Fan blade angle
 fixed
 Tunnel Pres. : 1 – 100 × 10⁵ Pa
 Tunnel Temp. : ambient to 45° C

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Test Section

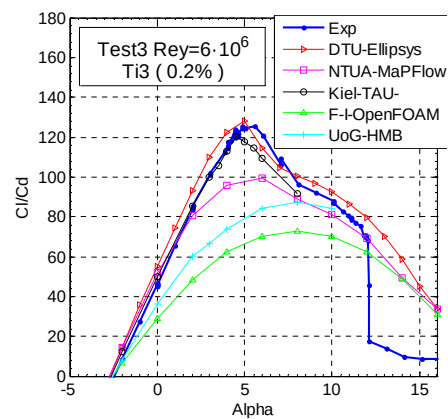
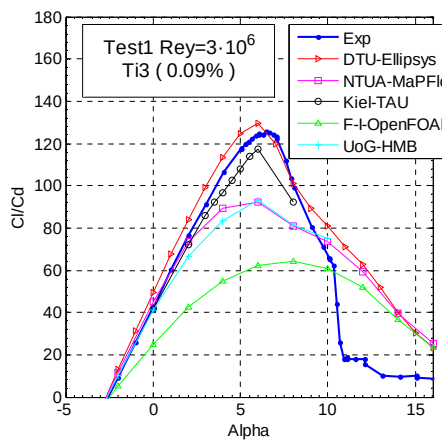


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Participants/Codes

			Test 1.	Test 2.	Test 3.	Test 4.	Test 5.	Test 6.	Test 7.
			Re=3mil	Re=6mil-1	Re=6mil-2	Re=9mil-1	Re=9mil-2	Re=12mil	Re=15mil
			12	34	67	34	67	67	60
			25.6	19	10	28.6	15	20	28.4
Full CFD	DTU/EllipSys	Fully turbulent							
		Transition	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	KIEL/TAU	Fully turbulent							
		Transition	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	NTUA/Mapflow	Fully turbulent							
		Transition	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	UoG/HMB	Fully turbulent							
		Transition	Yes		Yes				Yes
	Forwind-IWES/OpenFOAM	Fully turbulent	Yes	Yes	Yes	Yes	Yes	Yes	Yes
		Transition							
Panel Methods	USTUTT/XFOILvUSTUTT	Fully turbulent							
		Transition	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	ORE Catapult/XFOILv6.96	Fully turbulent							
		Transition	Yes	Yes	Yes	Yes	Yes	Yes	Yes

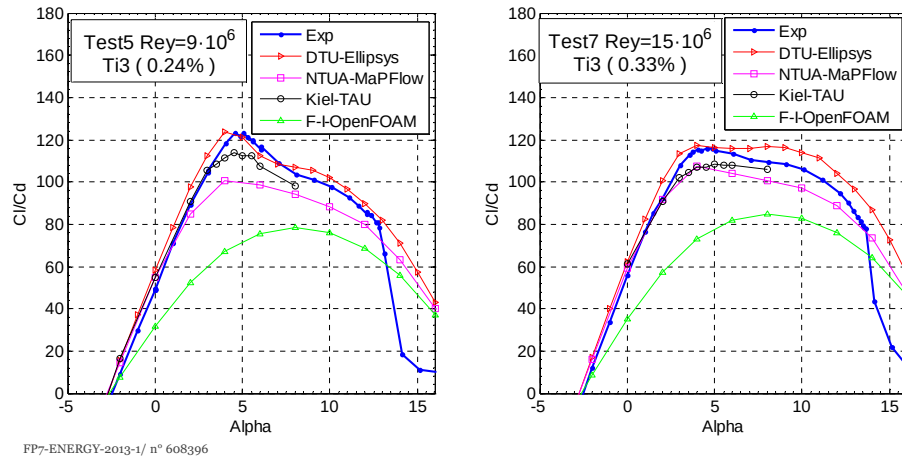
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DNW-HDG Full CFD calculations vs measurements
Effect in Blade Design parameter: Cl/Cd 

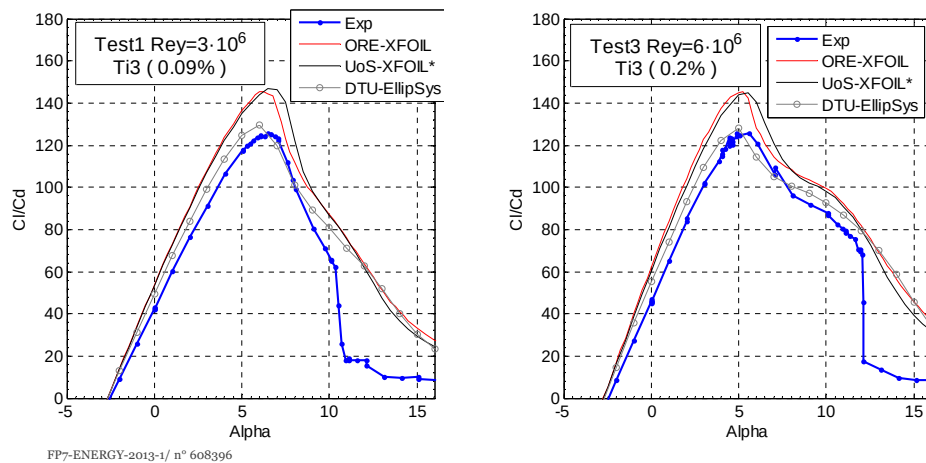
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DNW-HDG Full CFD calculations vs measurements
Effect in Blade Design parameter: Cl/Cd

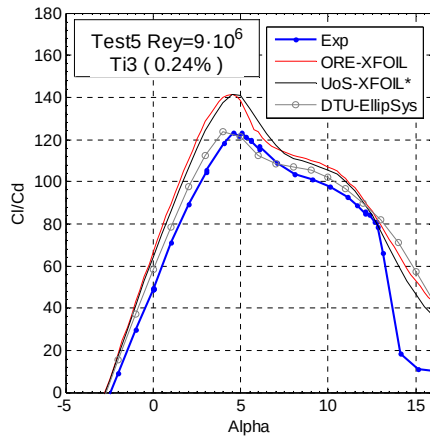


DNW-HDG Panel code calculations vs measurements
Effect in Blade Design parameter: Cl/Cd

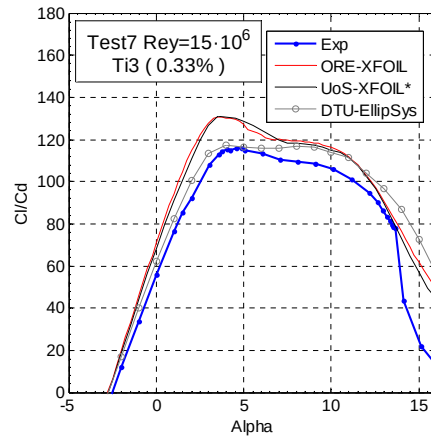




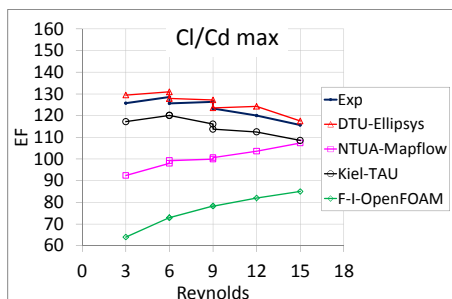
DNW-HDG Panel code calculations vs measurements Effect in Blade Design parameter: Cl/Cd



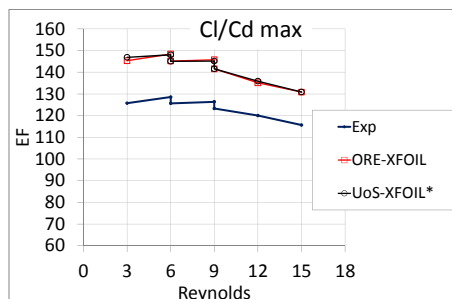
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Results Re effects in Cl/Cd trends



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List of Contents

- ☐ Introduction into the project
- ☐ Design of AVATAR Reference Wind Turbine ¹⁾
- ☐ Aerodynamics at high Reynolds numbers
 - Results from a blind test on airfoil measurements taken in the pressurized DNW-HDG tunnel ²⁾
- ☐ Aerodynamics of flow devices
 - *Results from measurements and calculations on vg's ³⁾*

1) Acknowledgement G. Sieros, M. Stettner

2) Acknowledgement O. Ceyhan, O. Pires

3) Acknowledgement M. Manolesos, J. Prospathopoulos, S. Voutsinas

And many others

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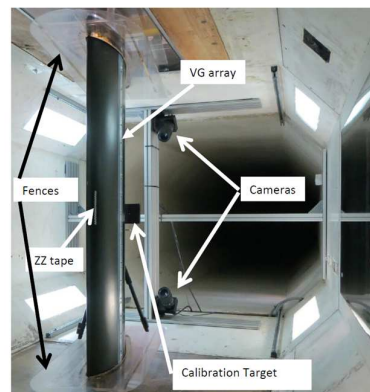
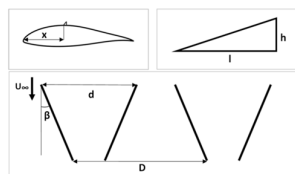
25-11-2016



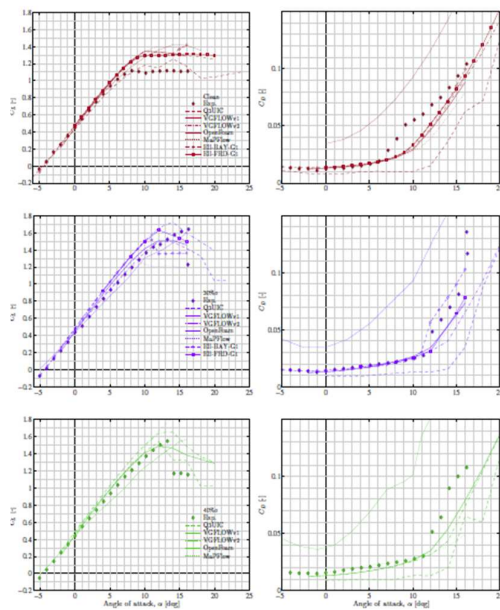
NTUA Measurements on vortex generators

Manolesos, M. and Prospathopoulos, J, S. Voutsinas

- NTUA measurements on NTUA 18% airfoil
- $Re = 0.87 \cdot 10^6$
- Pressure measurements, wake rig and PIV



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Comparison calculations and NTUA measured lift and drag polars with and without vg's

Top: No vg's

Middle: vg's at 30% chord

Bottom: vg's at 40% chord

Observations:

- Lift: Stall delay
- Drag: penalty at attached flow
- Relative effects of vg's are assessed through KPI's ¹⁾ and predicted reasonably well but there is spread in the clean conditions

Manolesos, M. and Prospathopoulos, J. (2015).
CFD and experimental database of
flow devices, comparison. Technical Report February.



Summary

- AVATAR is an EU FP7 projects which aims to validate, improve and calibrate aerodynamic models for 10MW+ turbines with and without flow devices and with and without aero-elastic effects
- Several (wind tunnel) measurements have been taken which have helped to validate and improve (sub) models relevant for 10MW+ turbines, e.g.
 - Measurements in a pressurized wind tunnel
 - Measurements taken at controlled turbulence
 - Measurements taken with flow devices
- Models are also evaluated on 2 Reference Wind Turbines
 - AVATAR low induction turbine with special aerodynamic challenges
 - INNWIND.EU conventional induction turbine
- The amount of results is far too much to present in 20 minutes
 - GOTO <http://www.eera-avtar.eu/publications-results-and-links/> !!!!!
- Note: Results from aero-elastic calculations with BEM and LL/FVW will be shown by Koen

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Consortium

Coordinator:



Partners in alphabetical order:



POLITECNICO DI MILANO



University of Stuttgart
Germany

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